



Arab Republic of Egypt
Ministry of Education and
Technical Education
Book Sector

Mathematics

For Primary 3

Second Term



Egyptian Knowledge Bank
بنك المعرفة المصري



غير مصرح بتداول هذا الكتاب خارج وزارة التربية والتعليم

2017-2018



Arab Republic of Egypt
Ministry of Education and
Technical Education
Book Sector

Mathematics

For Primary 3
Second Term

Authors:

Dr. Fayez Mourad Mina

Dr. Jean Michel Hanna



2017-2018

غير مصرح بتداول هذا الكتاب خارج وزارة التربية والتعليم

Revised by

MR/ Hussein Mahmoud Hussein

Consultant of Maths

Introduction

A foreword to teachers and parents

Dear teachers and parents,

We are pleased to present you with this book as part of a developed chain of mathematics textbooks. For maximum benefit, please note the following:

- 1_ Before solving the verbal problems, please read them out carefully to your pupils and make sure they are understood.
- 2_ There are multiple correct answers to some of the problems. It is sufficient for your pupils to mention only one or some according to what is required in the problem. It is with these types of questions that we hope to develop our pupils' creativity.
- 3_ An attempt has been made to remove barriers between mathematics and other areas of knowledge on the one hand, and practical life on the other hand, according to what has come to be known as "curriculum integration". Then the best time to start is the primary stage. Therefore, it is expected that every single detail in the book will be given attention and care even if it does not belong to "mathematics" in the narrow sense of the word.
- 4_ Some affective aims have been included in this curriculum. This is achieved by forming attitudes towards some social issues (such as the population issue) besides developing appreciation and interests towards the study of mathematics. Therefore, required discussions, comments, and other like response should not be ignored under the pretext that they are not included in school tests.
- 5_ It is not only the customary standards of education in Egypt that have been given apparent attention, but also modern trends in the teaching of mathematics. Among these are presenting comprehensive knowledge of numbers before details pertaining to place value and performing arithmetic operations.
- 6_ In the course of designing this book, circumstances of Egyptian schools have been taken into consideration. Hence the use of measuring tools and the performance of practical experiments have been kept to a minimum.
- 7_ There are activities and exercises at the end of each unit. The exercises are typical of the preplanned output of each unit. The activities, however, might sometimes exceed the contents of the unit with the purpose of reviving extracurricular activities in mathematics. These, in general, support the output of the unit and can be viewed as enrichment activities at the same time.

May God guide us all to what is in the best interest of our beloved country

The authors

Table of Contents

Unit 1: Multiplication and division	5
Lesson 1 : Multiplying by 10	6
Lesson 2 : Multiplying by 100	10
Lesson 3 : Multiplying by 1000	13
Lesson 4 : Multiplying a 2 - digit Number or more by a 1 - digit Number	16
Lesson 5 : Even Numbers and Odd Numbers	20
Lesson 6 : Dividing a Number by a 1- digit Number	22
Exercises on unit 1	25
Unit 1 Activities	29
Unit 2: Geometry	31
Lesson 1 : The Perimeter	32
Lesson 2 : The Area	39
Exercises on unit 2	42
Unit 2 Activities	44
Unit 3: Fractions	45
Lesson 1 : The Meaning and Reading of Fractions	46
Lesson 2 : Equal Fractions	49
Lesson 3 : Comparing and ordering fractions	52
Lesson 4 : Adding and subtracting the fractions	55
Exercises on unit 3	57
Unit 3 Activities	59
Unit 4: Measurement	61
Lesson 1 : Measuring Temperature	62
Lesson 2 : Measuring Length	65
Lesson 3 : Measuring Weight	67
Lesson 4 : Measuring Time	69
Exercises on unit 4	74
Unit 4 Activities	76
Unit 5: Statistics and Probability	77
Lesson 1 : Representing Data	78
Lesson 2 : Probability	83
Exercises on unit 5	87
Unit 5 Activities	88
General Exercises on units	90
Model tests	112

Unit1

Multiplication and division



Multiplying $\times 10$

(1) In the opposite figure:

The number of stars

$$= 10 + 10$$

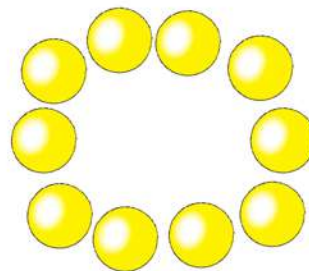
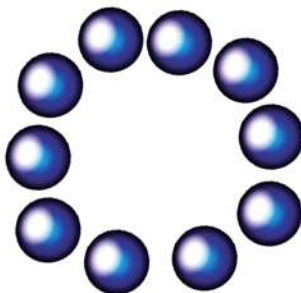
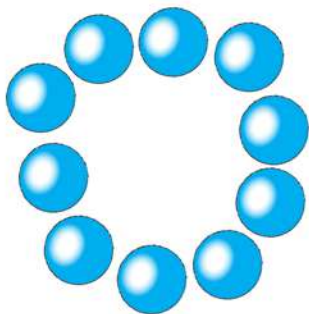
$$= 10 \times \dots = 20$$



(2) In the opposite figure :

The number of balls = $10 + \dots + \dots$

$$= \dots \times 3 = \dots$$



(3) Complete :

$$10 \times 4 = 10 + \dots + \dots + \dots = \dots$$

$$10 \times 5 = \dots + \dots + \dots + \dots + \dots = \dots$$

$$10 \times 6 = \dots + \dots + \dots + \dots + \dots + \dots = \dots$$

(4) Note the direction of the arrows, then complete the following table:

	1	2	3	4	5	6	7	8	9	10
	10		30							

(5) Complete:

$$10 \times 10 = \text{ten tens} = 6 \text{ tens} + \dots \text{ tens}$$

$$= 60 + \dots = \dots$$

$$7 \times 10 = 10 \times \dots = \dots \text{ tens} = 4 \text{ tens} + \dots \text{ tens}$$

$$= \dots + \dots = \dots$$

N.B.

$$2 \times 10 = 20, \quad 3 \times 10 = 30, \quad 4 \times 10 = 40, \quad \dots \text{and so on.}$$

For example : $7 \times 10 = 10 \times 7 = 70,$

$$15 \times 10 = 10 \times 15 = 150,$$

$$68 \times 10 = 10 \times 68 = 680 \text{ and so on.}$$

Exercises

(1) Find the result :

(a) $6 \times 10 = \dots\dots\dots$

(b) $9 \times 10 = \dots\dots\dots$

(c) $4 \times 10 = \dots\dots\dots$

(d) $2 \times 4 \times 10 = \dots\dots\dots$

(e) $3 \times 6 \times 10 = \dots\dots\dots$

(f) $3 \times 2 \times 10 = \dots\dots\dots$

(g) $15 \times 10 = \dots\dots\dots$

(h) $24 \times 10 = \dots\dots\dots$

(i) $2 \times 8 \times 10 = \dots\dots\dots$

(j) $10 \times 10 = \dots\dots\dots$

(2) Complete each of the following :

(a) $\dots\dots \times 10 = 30$

(b) $10 \times \dots\dots = 60$

(c) $\dots\dots \times 10 = 120$

(d) $10 \times \dots\dots = 180$

(e) $\dots\dots \times 10 = 90$

(f) $\dots\dots \times 10 = 160$

(g) $10 \times \dots\dots = 120$

(h) $5 \text{ tens} + 3 \text{ tens} = \dots\dots \times 10$

(3) Complete using the suitable sign ($>$, $<$, $=$)

(a) 5×10

10×6

(b) $2 \times 3 \times 10$

6×10

(c) 9 tens

9×9

(d) 10×15

$1 \times 5 \times 10$

(e) $3 \text{ tens} + 7 \text{ tens}$

73×10

(4) Complete:

(a) $8 \times 10 = 10 \times \dots = \dots$

(b) $\dots \times 10 = 10 \times \dots = 90$

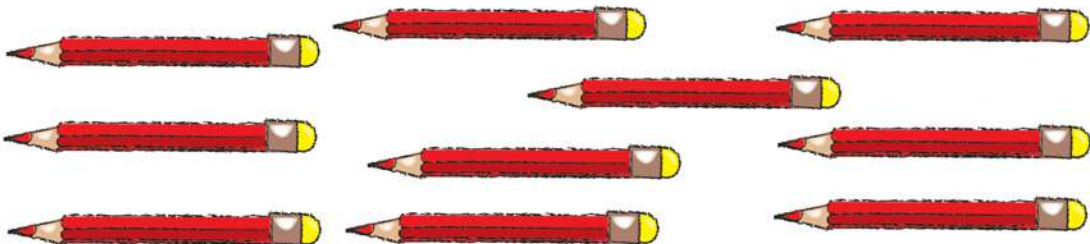
(c) $10 \times 2 \times 3 = 10 \times \dots = \dots$

(d) $10 \times 12 = 12 \times \dots = \dots$

(e) $5 \times 7 \times 10 = \dots \times 10 = \dots$

(f) $\dots \times 10 = 10 \times \dots = 530$

(5) How many pounds should Raouf pay to buy 10 pencils for 2 pounds each.



(6) Noura bought 10 notebooks and 2 books. How much did Noura pay if you know that the price of the notebook is three pounds and the price of the book is ten pounds?

.....

.....

.....

Multiplying $\times 100$

Complete:

$$100 \times 2 = 100 + 100 = \dots\dots\dots$$

$$100 \times 4 = 100 + \dots\dots\dots + \dots\dots\dots + \dots\dots\dots = \dots\dots\dots$$

$$100 \times 7 = 100 + \dots\dots\dots = \dots\dots\dots$$

We can continue in the same way until we reach $100 \times 10 = 1000$

We can also consider that $= 100 \times 10 =$ ten hundreds

$=$ five hundreds + five hundreds

$= 500 + 500 = 1000$

And also $100 \times 10 =$ ten hundreds

$=$ seven hundreds + three hundreds

$= 700 + \dots\dots\dots = \dots\dots\dots$ And so on

This can be arranged in the following two tables:

Complete:

$100 \times 0 = \dots\dots\dots$
$100 \times 1 = \dots\dots\dots$
$100 \times 2 = \dots\dots\dots$
$100 \times 3 = \dots\dots\dots$
$100 \times 4 = \dots\dots\dots$
$100 \times 5 = \dots\dots\dots$
$100 \times 6 = \dots\dots\dots$
$100 \times 7 = \dots\dots\dots$
$100 \times 8 = \dots\dots\dots$
$100 \times 9 = \dots\dots\dots$
$100 \times 10 = \dots\dots\dots$

$0 \times 100 = \dots\dots\dots$
$1 \times 100 = \dots\dots\dots$
$2 \times 100 = \dots\dots\dots$
$3 \times 100 = \dots\dots\dots$
$4 \times 100 = \dots\dots\dots$
$5 \times 100 = \dots\dots\dots$
$6 \times 100 = \dots\dots\dots$
$7 \times 100 = \dots\dots\dots$
$8 \times 100 = \dots\dots\dots$
$9 \times 100 = \dots\dots\dots$
$10 \times 100 = \dots\dots\dots$

N.B.

$2 \times 100 = 200$, $5 \times 100 = 500$, $9 \times 100 = 900$, .. and so on.
 For example: $7 \times 100 = 700$, $18 \times 100 = 1800$,
 $37 \times 100 = 3700$, and so on

Exercises

(1) Find the result of each of the following:

- | | |
|---|---|
| (a) $3 \times 100 = \dots\dots\dots$ | (b) $5 \times 100 = \dots\dots\dots$ |
| (c) $2 \times 3 \times 100 = \dots\dots\dots$ | (d) $4 \times 3 \times 100 = \dots\dots\dots$ |
| (e) $3 \times 3 \times 100 = \dots\dots\dots$ | (f) $10 \times 100 = \dots\dots\dots$ |

(2) Complete each of the following:

- | | |
|---|--|
| (a) $\dots\dots \times 100 = 900$ | (b) $100 \times \dots\dots = 500$ |
| (c) $\dots\dots \times 100 = 1200$ | (d) $100 \times \dots\dots = 1800$ |
| (e) $\dots\dots \times 3 \times 100 = 2700$ | (f) $24 \times \dots\dots = 2400$ |
| (g) $100 \times \dots\dots = 1000$ | (h) $30 \times 40 = 100 \times \dots\dots = \dots\dots$ |
| (i) $20 \times 90 = 100 \times \dots\dots = \dots\dots$ | (j) $40 \times 60 = \dots\dots \times \dots\dots = \dots\dots$ |
| (k) $10 \times 150 = \dots\dots \times \dots\dots = \dots\dots$ | |

(3) Complete:

$$6 \times 100 = 100 \times \dots = \dots, \quad \dots \times 100 = 100 \times \dots = 700$$

$$2 \times 4 \times 100 = \dots \times 100 = \dots, \quad 100 \times 28 = \dots \times 100 = \dots$$

$$100 \times 5 \times 7 = 100 \times \dots = \dots, \quad \dots \times 100 = 100 \times \dots = 7400$$

(4) As you know, one metre = 100 centimetres , complete:

$$7 \text{ metres} = \dots \text{ centimetres} \quad \text{because } 7 \times 100 = \dots$$

$$9 \text{ metres} = \dots \text{ centimetres} \quad \text{because } \dots \times 100 = \dots$$

$$25 \text{ metres} = \dots \text{ centimetres} \quad \text{because } \dots \times 100 = \dots$$

(5) If you know that Adel's father saves 100 pounds a month regularly, complete:

$$\begin{aligned} \text{What he saves in 8 months} &= \dots \text{ pounds} \\ &\text{because } 8 \times 100 = \dots \end{aligned}$$

$$\begin{aligned} \text{What he saves in 10 months} &= \dots \text{ pounds} \\ &\text{because } 10 \times 100 = \dots \end{aligned}$$

$$\begin{aligned} \text{What he saves in a whole year} &= \dots \text{ pounds} \\ &\text{because } \dots \times \dots = \dots \end{aligned}$$

Lesson 3

Multiplying $\times 1000$

Complete:

$$1000 \times 3 = 1000 + 1000 + 1000 = \dots\dots\dots$$

$$1000 \times 5 = 1000 + \dots\dots\dots + \dots\dots\dots + \dots\dots\dots + \dots\dots\dots = \dots\dots\dots$$

We can continue in the same way until we reach $1000 \times 10 = 10000$

We can also consider that :

$$1000 \times 10 = \text{ten thousands}$$

$$= \text{seven thousands} + \text{three thousands}$$

$$= 7000 + 3000 = 10000$$

Complete: $10000 \times 10 = \text{ten thousands}$

$$= 4 \text{ thousands} + \dots\dots\dots \text{ thousands}$$

$$= \dots\dots\dots + \dots\dots\dots = \dots\dots\dots$$

Notice the following :

$$8 \times 1000 = 8000$$

$$29 \times 1000 = 29000$$

$$47 \times 1000 = 47000$$

Exercises

(1) Find the result:

(a) $3 \times 1000 = \dots\dots\dots$

(b) $1000 \times 6 = \dots\dots\dots$

(c) $2 \times 2 \times 1000 = \dots\dots\dots$

(d) $1000 \times 3 \times 6 = \dots\dots\dots$

(e) $45 \times 1000 = \dots\dots\dots$

(f) $1000 \times 126 = \dots\dots\dots$

(g) $7 \times 9 \times 1000 = \dots\dots\dots$

(h) $6 \times 6 \times 1000 = \dots\dots\dots$

(i) $300 \times 50 = \dots\dots\dots$

(j) $200 \times 80 = \dots\dots\dots$

(2) Complete:

$$7 \times 1000 = 1000 \times \dots = \dots$$

$$\dots \times 1000 = 1000 \times \dots = 9000$$

$$2 \times 4 \times 1000 = \dots \times 1000 = \dots$$

$$3 \times \dots \times 1000 = \dots \times 1000 = 6000$$

$$2 \times \dots \times 1000 = 1000 \times \dots = 10000$$

$$7 \times \dots \times 1000 = 1000 \times \dots = 56000$$

(3) Complete:

(a) $300 \times 40 = 1000 \times \dots = \dots$

(b) $8 \times 500 = \dots \times 1000 = \dots$

(c) $6 \times 2000 = \dots \times 1000 = \dots$

(d) $100 \times 170 = 1000 \times \dots = \dots$

(e) $\dots \times 10 = 52 \times 1000 = \dots$

(f) $\dots \times 100 = \dots \times 1000 = 7000$

(4) Complete using the suitable sign (> or < or =)

(a) 100×2		1000×2
(b) 6×100		$2 \times 3 \times 10$
(c) 18×10		8×100
(d) $2 \times 7 \times 100$		14×10
(e) 52×100		25×100
(f) seven hundreds		7×100

- | | | |
|---------------------|----------------------|-----------------|
| (g) 300 | <input type="text"/> | three thousands |
| (h) 6 metres | <input type="text"/> | 500 centimetres |
| (j) 30×400 | <input type="text"/> | 200×60 |

(5) Complete as in the example :

Example : $10 \times 500 = 100 \times 50$
 $= 1000 \times 5$
 $= 5000$

- (a) $100 \times 60 = 10 \times \dots = 1000 \times \dots = \dots$
 (b) $20 \times 600 = 1000 \times 3 \times \dots$
 (c) $30 \times 600 = 1000 \times 2 \times \dots$
 (d) $40 \times 500 = 1000 \times 2 \times \dots$

(6) The average profit of a shop is 1000 pounds a week. What do you expect the profit to be in:

- (a) Six weeks?
- (b) eight weeks?
- (c) twenty weeks?

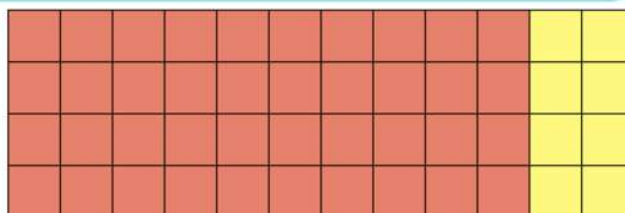
(7) An owner of a library bought 1000 notes, if the price of the note is 14 pounds. Find the total money did he pay

He paid = = pound

Multiplying a 2-digit Number or more by a 1-digit Number

Example (1) : Find the product 12×4 :

Consider 12×4 to be the number of the small squares that the opposite figure contains.



(Because the number of squares it contains is $12 + 12 + 12 + 12$)

$$\begin{aligned} \text{The number of small squares} &= 2 \times 4 + 10 \times 4 \\ &= 8 + 40 = 48 \end{aligned}$$

The solution can therefore written in the following form :

$$\begin{array}{r} 12 \\ \times 4 \\ \hline \end{array} \longrightarrow \begin{array}{r} 2 + 10 \\ \times 4 \\ \hline 8 + 40 \end{array}$$

This means that $12 \times 4 = 48$
also $4 \times 12 = 48$

Example (2): Find the product of 123×5 :

$$123 = 3 + 20 + 100$$

$$\begin{array}{r} 123 \\ \times 5 \\ \hline \end{array} \longrightarrow \begin{array}{r} 3 + 20 + 100 \\ \times 5 \\ \hline 15 + 100 + 500 = 615 \end{array}$$

This means that $123 \times 5 = 615$
also $5 \times 123 = 615$

Multiplying a 2-digit Number or more by a 1-digit Number

Example (3) : Find the product of 265×3

$$265 = 5 + 60 + 200$$

$$\begin{array}{r} 265 \\ \times 3 \\ \hline \end{array}$$



$$\begin{array}{r} 5 + 60 + 200 \\ \times 3 \\ \hline \end{array}$$

.....

$$15 + 180 + 600 = 15 + 80 + 700 = 795$$

This means that $265 \times 3 = 795$

also $3 \times 265 = 795$

(1) Complete as the example :

(a)	$\begin{array}{r} 1 \\ 46 \\ \times 3 \\ \hline 138 \end{array}$	$\begin{array}{r} 53 \\ \times 4 \\ \hline \end{array}$	$\begin{array}{r} 72 \\ \times 6 \\ \hline \end{array}$	$\begin{array}{r} 48 \\ \times 7 \\ \hline \end{array}$
(b)	$\begin{array}{r} 22 \\ 157 \\ \times 4 \\ \hline 628 \end{array}$	$\begin{array}{r} 236 \\ \times 4 \\ \hline \end{array}$	$\begin{array}{r} 348 \\ \times 6 \\ \hline \end{array}$	$\begin{array}{r} 635 \\ \times 5 \\ \hline \end{array}$
(c)	$\begin{array}{r} 412 \\ 1623 \\ \times 7 \\ \hline 11361 \end{array}$	$\begin{array}{r} 2347 \\ \times 5 \\ \hline \end{array}$	$\begin{array}{r} 1842 \\ \times 8 \\ \hline \end{array}$	$\begin{array}{r} 2432 \\ \times 9 \\ \hline \end{array}$

(2) Multiply :

$\begin{array}{r} 27 \\ \times 8 \\ \hline \end{array}$	$\begin{array}{r} 56 \\ \times 3 \\ \hline \end{array}$	$\begin{array}{r} 128 \\ \times 6 \\ \hline \end{array}$	$\begin{array}{r} 316 \\ \times 9 \\ \hline \end{array}$
$\begin{array}{r} 2234 \\ \times 2 \\ \hline \end{array}$	$\begin{array}{r} 6234 \\ \times 3 \\ \hline \end{array}$	$\begin{array}{r} 5489 \\ \times 4 \\ \hline \end{array}$	$\begin{array}{r} 6743 \\ \times 9 \\ \hline \end{array}$

(3) Notice that:

The vertical method of multiplication is the same as the horizontal method

$$24 \times 3 = 72$$

Find the result :

$$\begin{array}{r} 24 \\ \times 3 \\ \hline 72 \end{array}$$

(a) $43 \times 5 = \dots\dots\dots$

(b) $76 \times 4 = \dots\dots\dots$

(c) $286 \times 6 = \dots\dots\dots$

(d) $304 \times 2 = \dots\dots\dots$

(e) $2261 \times 7 = \dots\dots\dots$

(f) $5092 \times 4 = \dots\dots\dots$

(g) $3002 \times 5 = \dots\dots\dots$

(h) $555 \times 7 = \dots\dots\dots$

(i) $2211 \times 9 = \dots\dots\dots$

(j) $5678 \times 4 = \dots\dots\dots$

(4) Mostafa bought 3 metres of cloth to have a suit made. How much did Mostafa pay if the price of 1 metre is 89 pounds?

The price of the cloth = $\dots\dots\dots \times \dots\dots\dots = \dots\dots\dots$ pounds

(5) Mahmoud went to the market. He had twenty pounds. He bought 2 kgms of tomatoes for 3 pounds a kilo , 1 kilogram of cucumber for 2 pounds and 3 kgms of oranges for 2 pounds a kilo. How many pounds did mahmoud have when he left the market ? $\dots\dots\dots$



Multiplying a 2-digit Number or more by a 1-digit Number

(6) Complete as the first, second example:

$$\begin{array}{r} 645 \\ \times 7 \\ \hline 4515 \end{array}$$

$$\begin{array}{r} 2368 \\ \times 9 \\ \hline 21312 \end{array}$$

$$\begin{array}{r} 5 \square 86 \\ \times 6 \\ \hline 347\square\square \end{array}$$

$$\begin{array}{r} 6\square 57 \\ \times 6 \\ \hline 3\square 7\square 2 \end{array}$$

$$\square 98 \times 5 = 3490$$

$$5 \times 4 \square 3 = 231\square$$

(7) Complete in the same sequence :

5, 25, 125, 625,,

10, 100, 10000,,

6, 36, 216,,

(8) Complete using one of the signs <, =, or > :

$$286 \times 3 \quad \square \quad 200 \times 4 \quad , \quad 1234 \times 9 \quad \square \quad 3702 \times 3$$

$$6594 \times 6 \quad \square \quad 7912 \times 5 \quad , \quad 2465 \times 7 \quad \square \quad 2875 \times 6$$

$$3698 \times 8 \quad \square \quad 4227 \times 7 \quad , \quad 4689 \times 10 \quad \square \quad 9378 \times 5$$

(9) Notice, then put (✓) in front of the correct answer and (X) in front of the uncorrect answer, give a reason :

(a) $264 \times 2 = 548$ (X), because the number in the tens digit is false

(b) $2 \times 3 \times 715 = 4290$ (✓), because the product gives the final answer

(c) $2708 \times 7 = 118956$ (), because

(d) $9 \times 6124 = 55116$ (), because

(e) $2 \times 5 \times 5756 = 507560$ (), because

Even Numbers and odd Numbers

(1) In the physical education period groups were formed in different games as follows :

Football : 14 pupils.

Basketball : 10 pupils .

Volleyball : 12 pupils.

Aerobics : 9 pupils.

The teacher asked the groups to stand in rows with 2 adjacent pupils in each row:



Complete:

Football : $14 \div 2 = \dots\dots\dots$ and $\dots\dots\dots$ remain.

Therefore the number of rows = $\dots\dots\dots$

Basketball : $10 \div 2 = \dots\dots\dots$ and $\dots\dots\dots$ remain.

Therefore the number of rows = $\dots\dots\dots$

Volleyball : $12 \div 2 = \dots\dots\dots$ and $\dots\dots\dots$ remain.

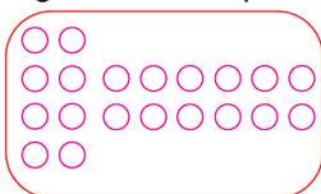
Therefore the number of rows = $\dots\dots\dots$

Aerobics : $9 \div 2 = \dots\dots\dots$ and $\dots\dots\dots$ remain.

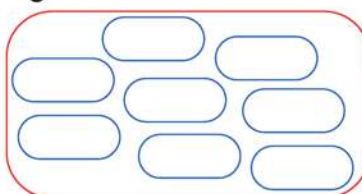
Therefore the number of rows = $\dots\dots\dots$

From this notice that in each of the first three groups each pupil could find a classmate to stand by to form a row. No one was left alone. In this case the whole number of pupils (that could be divided into pairs with no one remaining alone) is called an even number. As for the last group, we find that each pupil found another pupil to stand beside him, except for one who remained alone. The number in this case is called an odd number.

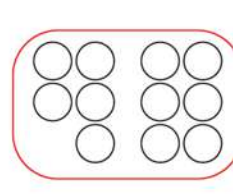
(2) In each of the following figures, we tried to form pairs. Notice each figure and complete using either “even” or “odd”



an $\dots\dots\dots$ number



an $\dots\dots\dots$ number



an $\dots\dots\dots$ number

(3) The even numbers are 0, 2, 4, 6, 8, 10, 12, .. or any number its unit digit 0, 2, 4, 6 or 8

Examples : 50, 172, 84, 436, 2008

The odd number are : 1, 3, 5, 7, 9, 11,

or any number its unit digit 1, 3, 5, 7 or 9

Examples : 61, 203, 95, 1237, 549

(4) Underline the even numbers :

4069, 17352, 74, 2003, 3002

(5) Underline the odd numbers :

174, 5103, 1035, 2406, 46

(6) Complete each of the following :

(a) Any even number + 1 = number

(b) Any even number - 1 = number

(c) Any odd number + 1 = number

(d) Any odd number - 1 = number

(e) Any even number + 2 = number

(f) Any odd number + 2 = number

(g) The even number just after the number 306 is the number.....

(h) The odd number just before the number 2751 is the number.....

Dividing a Number by a 1-digit Number

(1) In one of the libraries 484 books are to be distributed among four shelves equally. How many books should be put on each shelf?

We know that number of book on each shelf will be the quotient of the division operation $484 \div 4$, to find this quotient:

Consider that: $484 = 400 + 80 + 4$

$= 4 \text{ hundreds} + 8 \text{ tens} + 4 \text{ units}$

$4 \text{ hundreds} \div 4 = \text{one hundred} = 100$

$8 \text{ tens} \div 4 = \text{two tens} = 20$

$4 \text{ units} \div 4 = \text{one unit} = 1$

Therefore the number of books on each shelf $= 100 + 20 + 1 = 121$ books

Therefore $484 \div 4 = 121$



(2) Complete as in the example :

Example

$$396 \div 3$$

$$300 \div 3 = 100$$

$$90 \div 3 = 30$$

$$6 \div 3 = 2$$

$$396 \div 3 = 132$$

$$284 \div 2$$

$$200 \div 2 = \dots\dots$$

$$80 \div 2 = \dots\dots$$

$$4 \div 2 = \dots\dots$$

$$284 \div 2 = \dots\dots$$

$$848 \div 4$$

$$800 \div \dots\dots = \dots\dots$$

$$\dots\dots \div \dots\dots = \dots\dots$$

$$\dots\dots \div \dots\dots = \dots\dots$$

$$848 \div 4 = \dots\dots$$

(3) Find the quotient of each of the following division operations:

$$963 \div 3 = \dots\dots\dots,$$

$$460 \div 2 = \dots\dots\dots$$

$$804 \div 4 = \dots\dots\dots,$$

$$6482 \div 2 = \dots\dots\dots$$

$$9300 \div 3 = \dots\dots\dots,$$

$$7070 \div 7 = \dots\dots\dots$$

Dividing a Number by a 1-digit Number

$624 \div 2 = \dots\dots\dots,$

$666 \div 6 = \dots\dots\dots$

$999 \div 9 = \dots\dots\dots,$

$800 \div 2 = \dots\dots\dots$

$4682 \div 2 = \dots\dots\dots,$

$5500 \div 5 = \dots\dots\dots$

$8044 \div 4 = \dots\dots\dots$

(4) Find the result of each of the following :

$$\begin{array}{r} 2 \overline{) 624} \end{array}$$

$$\begin{array}{r} 3 \overline{) 936} \end{array}$$

$$\begin{array}{r} 5 \overline{) 500} \end{array}$$

$$\begin{array}{r} 2 \overline{) 2084} \end{array}$$

$$\begin{array}{r} 4 \overline{) 1608} \end{array}$$

$$\begin{array}{r} 6 \overline{) 3660} \end{array}$$

$$\begin{array}{r} 7 \overline{) 1407} \end{array}$$

$$\begin{array}{r} 9 \overline{) 9009} \end{array}$$

$$\begin{array}{r} 4 \overline{) 248} \end{array}$$

(5) Complete each of the following :

(a) The quotient of 505 by 5 is

(b) The quotient of 246 by 2 is

(c) $\div 2 = 32$

(d) $\div 3 = 203$

(e) $\div 5 = 21$

(f) $\div 3 = 333$

Unit 1

- (6) A merchant wanted to put 626 pieces of candy in two packets so that each packet would contain the same number of pieces.

What is the number of pieces in each packet?

Complete:

$$626 \div 2 = \dots\dots\dots$$

The number of pieces in each packet =



- (7) Samia and Mariam's father distributed among them 226 pounds equally. What is the share of each one?

The share of each one = = pounds

- (8) A primary school has 9 classes with equal numbers in each. If the whole number of pupils is 450 pupils. How many pupils are there in each class?

The number of pupils in each class =

- (9) An equal number of children are vaccinated against polio in one Ministry of Health Clinics. If 328 children are vaccinated in 8 days, then how many children were vaccinated in 5 days.

Number of children vaccinated in one day = =

Exercisers

Unit 1

(1) Find the result :

- (a) $7 \times 10 = \dots\dots\dots$ (e) $7 \times 2 \times 5 = \dots\dots\dots$ (i) $45 \times 1000 = \dots\dots\dots$
 (b) $8 \times 100 = \dots\dots\dots$ (f) $2 \times 4 \times 100 = \dots\dots\dots$ (j) $20 \times 60 = \dots\dots\dots$
 (c) $9 \times 1000 = \dots\dots\dots$ (g) $3 \times 6 \times 10 = \dots\dots\dots$ (k) $40 \times 50 = \dots\dots\dots$
 (d) $10 \times 10 = \dots\dots\dots$ (h) $4 \times 3 \times 100 = \dots\dots\dots$ (l) $300 \times 40 = \dots\dots\dots$

(2) Find the result of each of the following :

(a)	$\begin{array}{r} 24 \\ \times 5 \\ \hline \end{array}$	(b)	$\begin{array}{r} 16 \\ \times 3 \\ \hline \end{array}$	(c)	$\begin{array}{r} 47 \\ \times 4 \\ \hline \end{array}$	(d)	$\begin{array}{r} 48 \\ \times 6 \\ \hline \end{array}$
(e)	$\begin{array}{r} 217 \\ \times 3 \\ \hline \end{array}$	(f)	$\begin{array}{r} 504 \\ \times 5 \\ \hline \end{array}$	(g)	$\begin{array}{r} 721 \\ \times 4 \\ \hline \end{array}$	(h)	$\begin{array}{r} 415 \\ \times 2 \\ \hline \end{array}$
(i)	$\begin{array}{r} 1215 \\ \times 4 \\ \hline \end{array}$	(j)	$\begin{array}{r} 6100 \\ \times 6 \\ \hline \end{array}$	(k)	$\begin{array}{r} 4017 \\ \times 7 \\ \hline \end{array}$	(l)	$\begin{array}{r} 8231 \\ \times 8 \\ \hline \end{array}$

Unit 1

(3) Find the result:

(a) $639 \div 3 = \dots\dots\dots$

(d) $550 \div 5 = \dots\dots\dots$

(g) $660 \div 6 = \dots\dots\dots$

(b) $460 \div 2 = \dots\dots\dots$

(e) $164 \div 4 = \dots\dots\dots$

(h) $497 \div 7 = \dots\dots\dots$

(c) $250 \div 5 = \dots\dots\dots$

(f) $488 \div 8 = \dots\dots\dots$

(i) $360 \div 9 = \dots\dots\dots$

(4) Find the result :

(a) $\begin{array}{r} 2 \overline{) 408} \end{array}$

(b) $\begin{array}{r} 5 \overline{) 5010} \end{array}$

(c) $\begin{array}{r} 4 \overline{) 488} \end{array}$

(d) $\begin{array}{r} 7 \overline{) 4207} \end{array}$

(e) $\begin{array}{r} 3 \overline{) 183} \end{array}$

(f) $\begin{array}{r} 3 \overline{) 366} \end{array}$

(5) Complete :

(a) $\dots\dots\dots \times 100 = 900$

(g) $45 \times \dots\dots\dots = 45000$

(b) $10 \times \dots\dots\dots = 90$

(h) $40 \times \dots\dots\dots = 4000$

(c) $100 \times \dots\dots\dots = 1000$

(i) $3 \times 300 = \dots\dots\dots \times 100 = \dots\dots\dots$

(d) $10 \times \dots\dots\dots = 160$

(j) $10 \times 6 \times \dots\dots\dots = 6 \times 1000$

(e) $100 \times 4 \times \dots\dots\dots = 2400$

(k) $20 \times 30 = 10 \times \dots\dots\dots$

(f) $1000 \times \dots\dots\dots = 126000$

(l) $40 \times 50 = 20 \times \dots\dots\dots$

(6) Put the suitable sign (> or < or =) :

(a) 515×5

$515 \div 5$

(b) $369 \div 3$

100×3

(c) $3 \times 4 \times 1000$

12×1000

(d) 20×70

14×100

(e) $62 \div 2$

$155 \div 5$

(f) $3752 \div 7$

3752×7

(7) Choose the correct answer from those between brackets:

- (a) Which of the following operations doesn't represent an even number
(3 hundreds + hundred or $30 \times 2 \times 4$ or $45 \div 5$)

- (b) If $135 \times 4 = 630$, then $630 \div 4 = \dots\dots\dots$
(120 or 125 or 135)

- (c) The number which multiplied by 3129 the result will be 3129 is
(0 or 1 or 10)

- (d) The number of the even numbers included between 10 and 20 is
(2 or 4 or 6)

(8) Mohamed has 20 banknote, one hundred pounds each, 7 banknote, 10 pounds each. What Mohamed has?

Mohamed has = = pounds

(9) If the price of one golden gram is 562 pounds, what is the price of 4 grams of this golden?

The price of 4 grams = = pounds.

(10) Complete in the same pattern :

(a) 2 , 20, 200, ,

(b) 215 , 430 , 860 , ,

Activities

Unit 1

(1) If you know that $7 \times 5 = 35$, $7 \times 6 = 42$, $7 \times 8 = 56$

use these equalities to complete :

$$7 \times 11 = \dots + \dots = \dots , \quad 7 \times 14 = \dots + \dots = \dots$$

$$7 \times 13 = \dots + \dots = \dots , \quad 7 \times 19 = \dots + \dots = \dots$$

(2) If you know that :

$$49 \times 7 = 343 , \quad 49 \times 30 = 1470$$

Complete:

$$49 \times 70 = \dots , \quad 49 \times 3 = \dots$$

$$49 \times 77 = \dots + \dots = \dots , \quad 49 \times 33 = \dots + \dots = \dots$$

$$49 \times 37 = \dots + \dots = \dots , \quad 49 \times 73 = \dots + \dots = \dots$$

(3) Check that the following equalities are correct : (check your answer by using the calculator) “you can use a calculator”

$$32 = 2 + 3 + (3 \times 9)$$

$$75 = 5 + 7 + (7 \times 9)$$

$$63 = 3 + 6 + (6 \times 9)$$

$$80 = 0 + 8 + (8 \times 9)$$

Try other number of your own, then search for the reason why this pattern is always correct. Complete the following equalities (in the same way) :

Unit 1

$$47 = 7 + 4 + (4 \times \dots)$$

$$68 = 8 + \dots + (6 \times \dots)$$

$$96 = \dots$$

$$84 = \dots$$

$$59 = \dots$$

(4) Show whether the answers of the following problems are reasonable or unreasonable and give a reason :

(a) $598 \times 7 = 3986$ () : because :

(b) $1779 \div 6 = 596$ () : because :

(c) $2594 \div 4 = 6378$ () : because :

(d) $285 \times 9 = 2565$ () : because :

Unit two

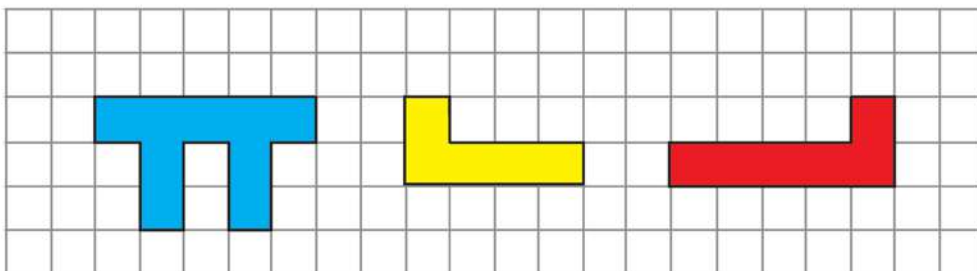
Geometry



The perimeter

You already know that the perimeter of any shape is the length of the line that outlines that shape and therefore:

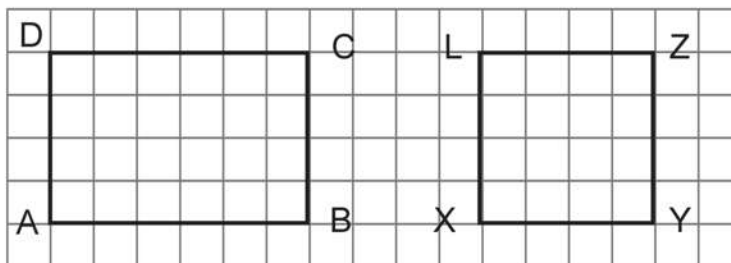
First: Calculate the perimeter of each of the following shapes (Consider the side length of the square as a unit length)



Perimeter = units

Perimeter = units

Perimeter = units



The Perimeter = unit length

The Perimeter = unit length

Second: Complete with the help of the opposite figure:

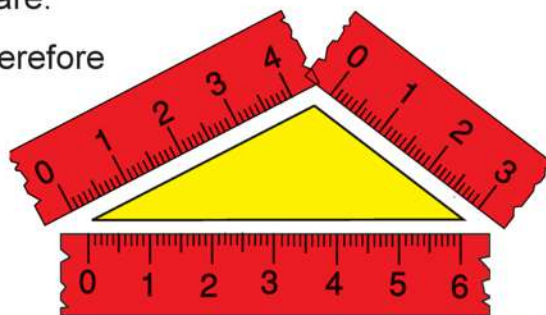
(a) the lengths of the triangle's sides are:

3 , 4 and 6 centimetres. Therefore

The triangle's perimeter:

$$= 3 + 4 + 6$$

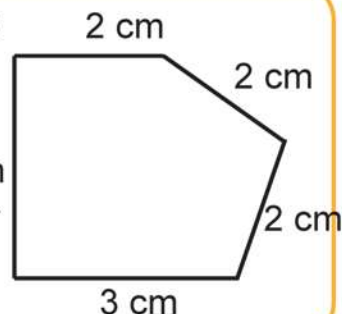
$$= 13 \text{ cm.}$$



(b) The lengths of the sides of the opposite figure are:

..... cm , cm , cm , cm , cm

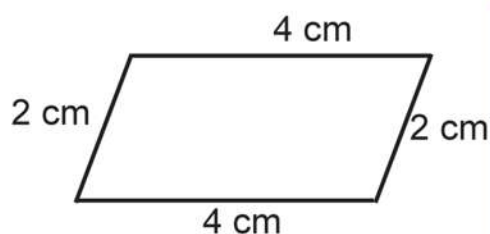
The Perimeter = + + + +
= cm



(c) The perimeter of the opposite figure

= + + +

= cm



From the previous, we deduce that

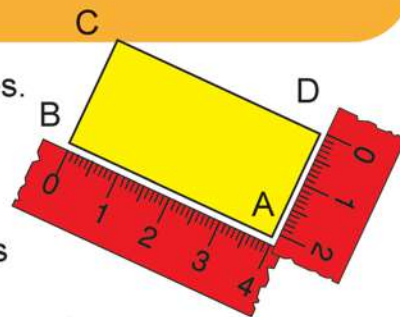
The perimeter of any polygon = The sum of its side lengths

(2) The opposite figure represents the rectangle A B C D with the help of the figure complete:

A B = centimetres . A D = centimetres.

Deduce the length of the other two sides.

C D = centimetres , B C = centimetres



The perimeter of the rectangle = + + + = centimetres

= $2 \times (..... +)$

= $2 \times = \text{ cm}$

Unit two

There fore:

The perimeter of the rectangle = (Length + width) $\times$ 2

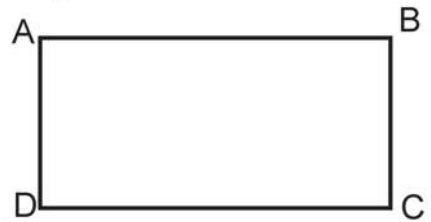
Notice that: each of

$\overline{AB}$, $\overline{DC}$ are called the length of the rectangle

$\overline{BC}$, $\overline{AD}$ are called the width of the rectangle,

where $AB = DC$, $BC = AD$

Each of the length, width of the rectangle are called the dimensions of the rectangle



Complete as in the example

The length in cm	The width in cm	The perimeter = (length + width) $\times$ 2
5cm	3cm	$(5 + 3) \times 2 = 16 \text{ cm}$
6	4	
7	3	
10	5	

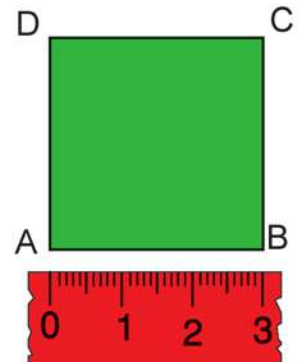
(3) The opposite figure represents square ABCD. With the help of the figure, complete:

$AB = \dots\dots\dots$ centimetres

Deduce the length of the remaining sides and calculate the square's perimeter.

The perimeter of the square = $\dots\dots\dots \times \dots\dots\dots$

= $\dots\dots\dots$ centimetres



There fore:

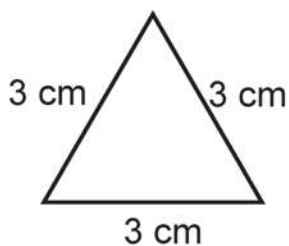
The perimeter of square = side length $\times$ 4

Complete as the example

Side length by cm	The perimeter of square = Side length $\times$ 4
Example 5	$5 \times 4 = 20$ cm
7	
4	
10	

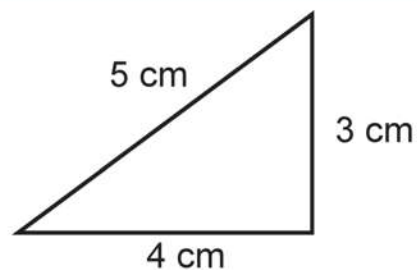
(4) Calculate the perimeter of each of the following figure:

(a)



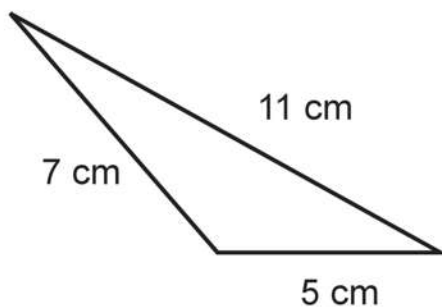
The perimeter = cm

(b)



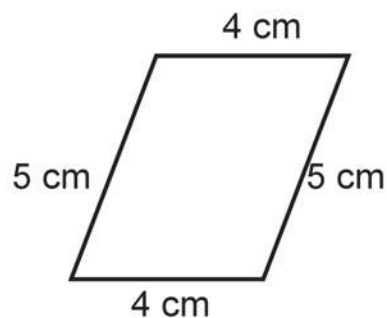
The perimeter = cm

(c)



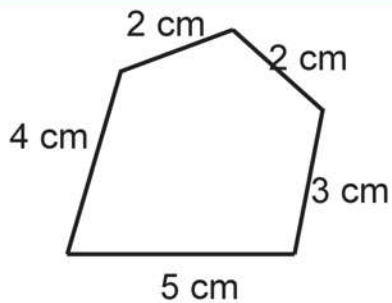
The perimeter = cm

(d)



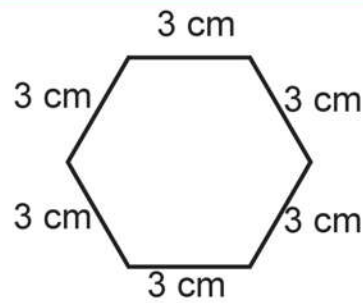
The perimeter = cm

(e)



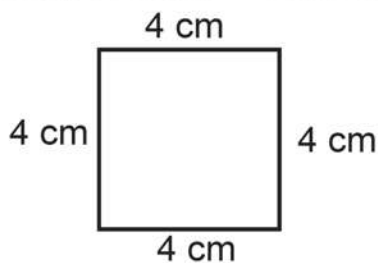
The perimeter = cm

(f)



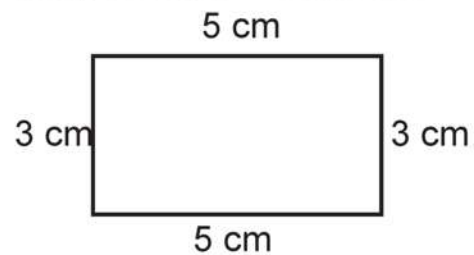
The perimeter = cm

(g)



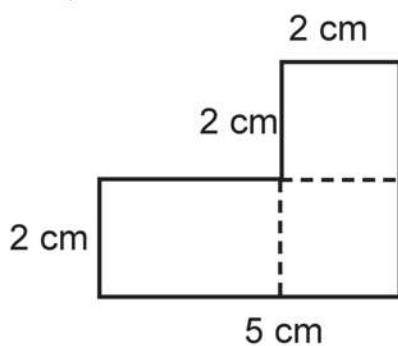
The perimeter = cm

(h)



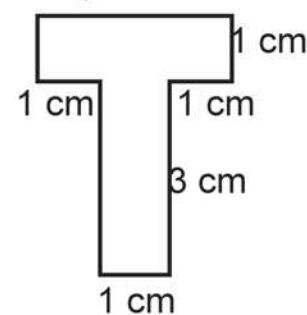
The perimeter = cm

(i)



The perimeter = cm

(k)

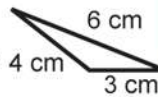


The perimeter = cm

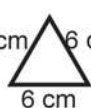
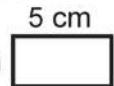
(5) Complete:

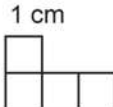
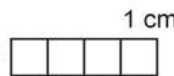
- (a) The perimeter of the square whose side length 5 cm = cm
- (b) The perimeter of the square whose side length 1 cm = cm
- (c) The perimeter of the rectangle whose length 8 cm, and its width 4 cm = cm
- (d) The perimeter of the rectangle whose dimensions are 16 cm, 10 cm = cm
- (e) The perimeter of the rectangle whose length 2 m and its width 150 cm = cm

(6) Put Suitable sign > or < or =

(a) The perimeter of square  the perimeter of triangle 

(b) The perimeter of rectangle  the perimeter of square 

(c) The perimeter of triangle  the perimeter of rectangle 

(d) The perimeter of figure  the perimeter of figure 

(7) Calculate the perimeter of a triangle whose sides are 4, 5 and 8 centimetres

The Perimeter = = cm

(8) The perimeter a triangular piece of land 200 metres, Find the length of its third side if you know that the sum of two of its sides is 140 metres.

The length of third side = = metres

(9) Calculate the perimeter of square of side length 7cm.

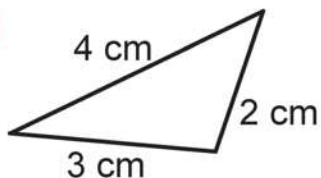
The perimeter of square = = cm

(10) Calculate the perimeter of rectangle of length 7cm and width 3cm.

The perimeter = = cm

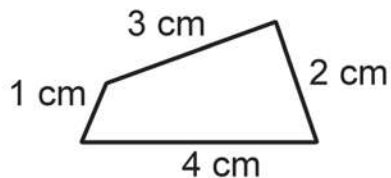
(11) Calculate the perimeter of each of the following shapes

(a)



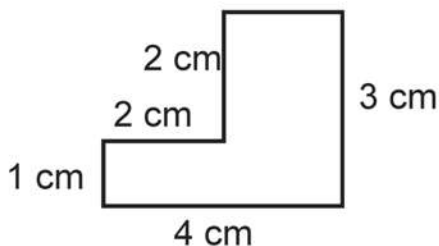
The perimeter = cm

(b)



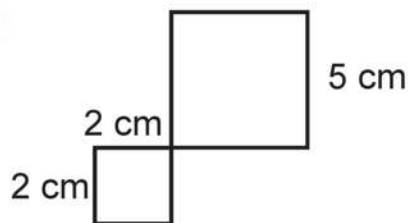
The perimeter = cm

(c)



The perimeter = cm

(d)



The perimeter = cm

(12) In the opposite shape

BCDE is a square where $AB = 3 \text{ cm}$, $AE = 5 \text{ cm}$ and $CD = 4 \text{ cm}$. Calculate

(a) The perimeter of square B C D E

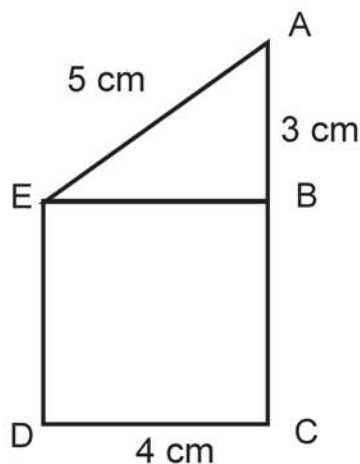
.....

(b) The perimeter of triangle A B E

.....

(c) The perimeter of the whole shape A C D E

.....



Lesson 2

The Area

- (1) To find out which of the following two shapes covers more space, find the number of small square that each shape consists of:

Complete:

Number of small squares in the first shape =

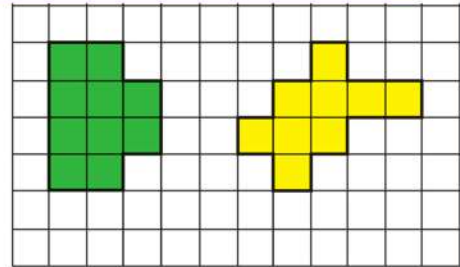
Number of small squares in the second shape =

Therefore, shape (.....) is the shape that covers more space.

In this case we say that the area of shape (.....) is greater than the area of shape (.....). In other words:

The area of a shape is defined by the number of units which consist that shape

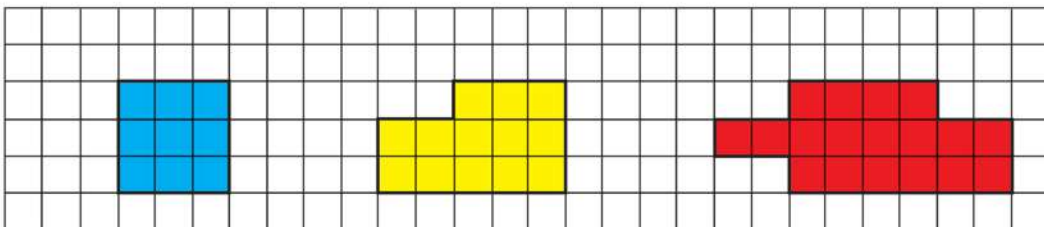
Then, an area of a shape depends on the used unit. If this unit is changed, the area of the shape is changed as well.



(1)

(2)

- (2) Calculate the area of each of the following shapes. Consider the small square a unit of area:

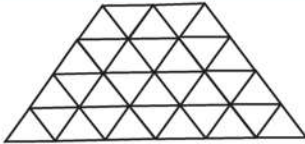


The area = 9 

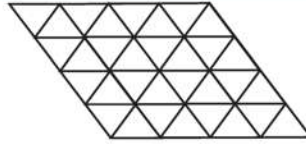
The area = 

The area = 

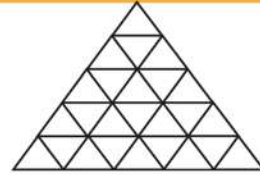
(3) Calculate the area of each of the following shapes, consider the small triangle  a unit of area.



the area = 



the area = 



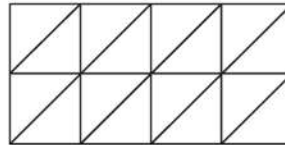
the area = 

(4) Write the area of the opposite figure according to the given unit.

the area = 

the area = 

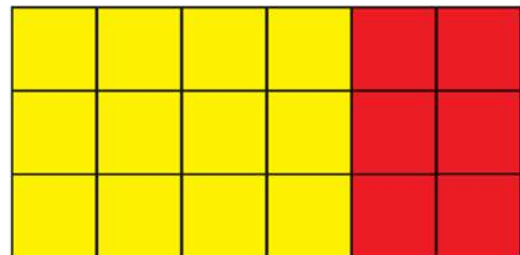
the area = 



(5) The opposite figure represents a large rectangle divided into two rectangles:

(a) Consider the length of the small square's side a unit of length and the area of the small square a unit of area then complete the following table:

	Perimeter	Area
The red rectangle		
The yellow rectangle		
The large rectangle		

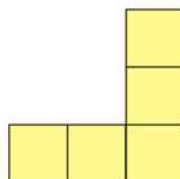
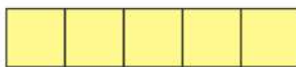
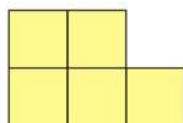


(b) Answer with Yes or No and explain why:

The perimeter of the large rectangle = the sum of the perimeter of the two rectangles
(Yes / No)

The area of the large rectangle = the sum of the areas of the two rectangles (Yes / No)

(6) In the following figures.



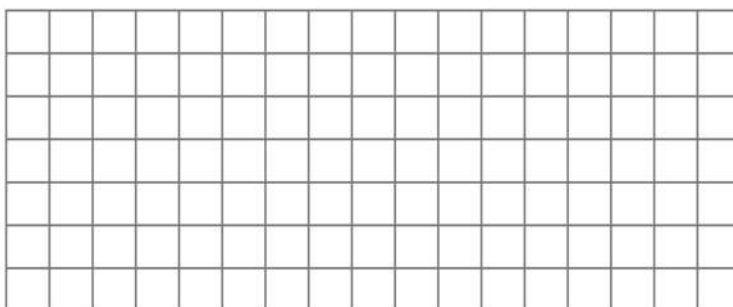
(a) Do the three opposite shapes have the same area? (Yes / No)

Why?

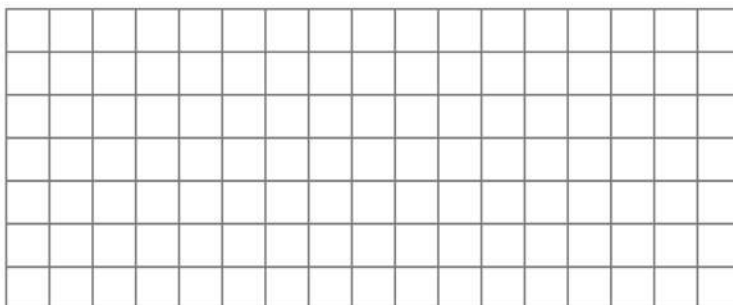
(b) search if they have the same perimeter? (Yes / No)

Why

(c) Draw two shapes that have the same area but not the same perimeter.



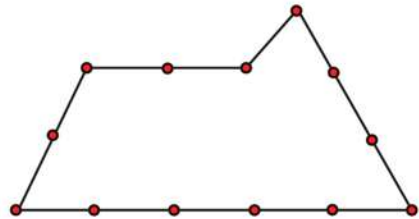
(d) Draw two shapes that have the same perimeter but not the same area.



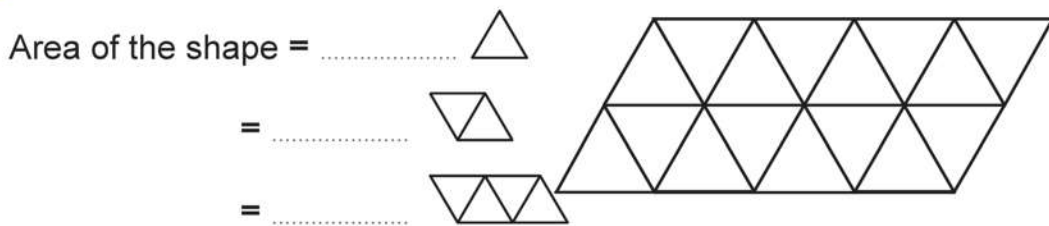
Exercises

Unit 2

- (1) Find the perimeter of this shape if you know that the distance between each two points is 1 centimetre long



- (2) Find the area of the opposite shape according to the given unit



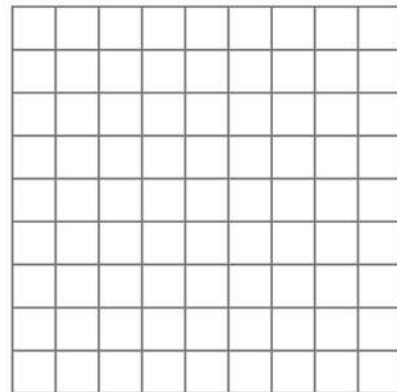
- (3) (a) Find the perimeter of a square whose side length is 3 cm.

the perimeter of the square = = cm

- (b) Find the perimeter of a triangle whose sides are 5 cm, 7 cm, and 10 cm.

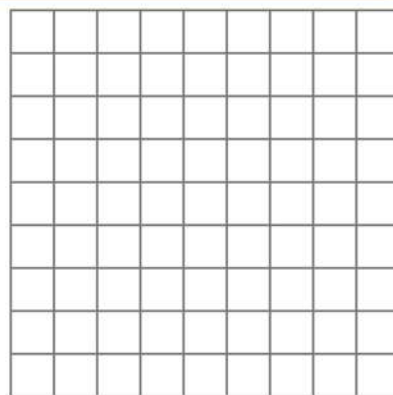
The perimeter of the triangle = = cm

- (4) On the opposite lattice: Draw a shape with a perimeter of 8 units of length.
(Consider the length of the small square's side a unit of length and its area a unit of area)

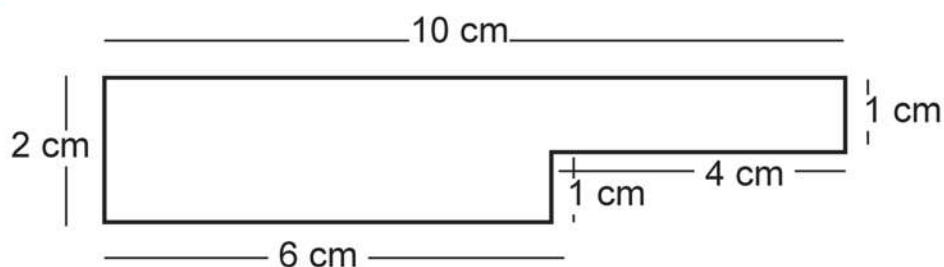


- (5) On the opposite lattice: Draw a shape with area of 8 square units.

(Consider the length of the small square's side a unit of length and its area a unit of area)



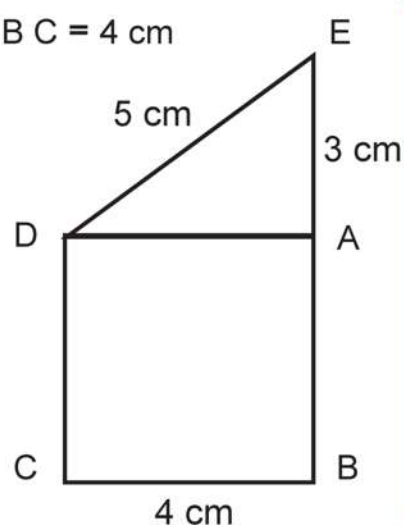
- (6) In the following figure calculate the perimeter of the figure in cm



- (7) In the opposite figure A B C D is square, B C = 4 cm

A E = 3 cm , E D = 5 cm.

Calculate its perimeter of the figure

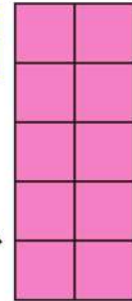
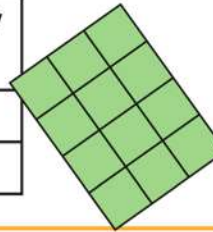
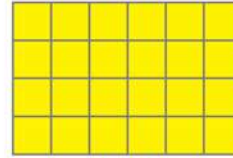


Activities

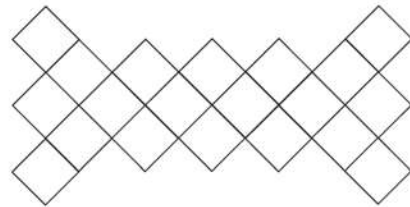
Unit 2

- (1) Find the perimeter and area of each of the rectangle in the opposite figure and complete the following table:

	The red rectangle	The green rectangle	The yellow rectangle
Perimeter			
Area			



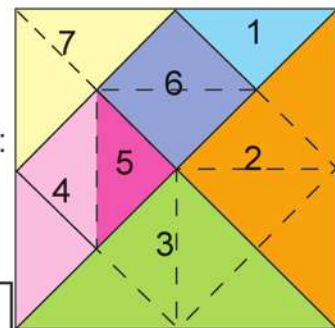
- (2) Find the perimeter and the area of the opposite shape according to the given unit:
 The perimeter = units
 (Consider the length of the small square's side a unit)



The area =  =  = 

- (3) The opposite shape represents a square divided into seven shapes numbered 1 to 7.
 Consider the area of shape (1) a unit of area and:
 (a) Find the area of the rest of the shapes and complete the following table:

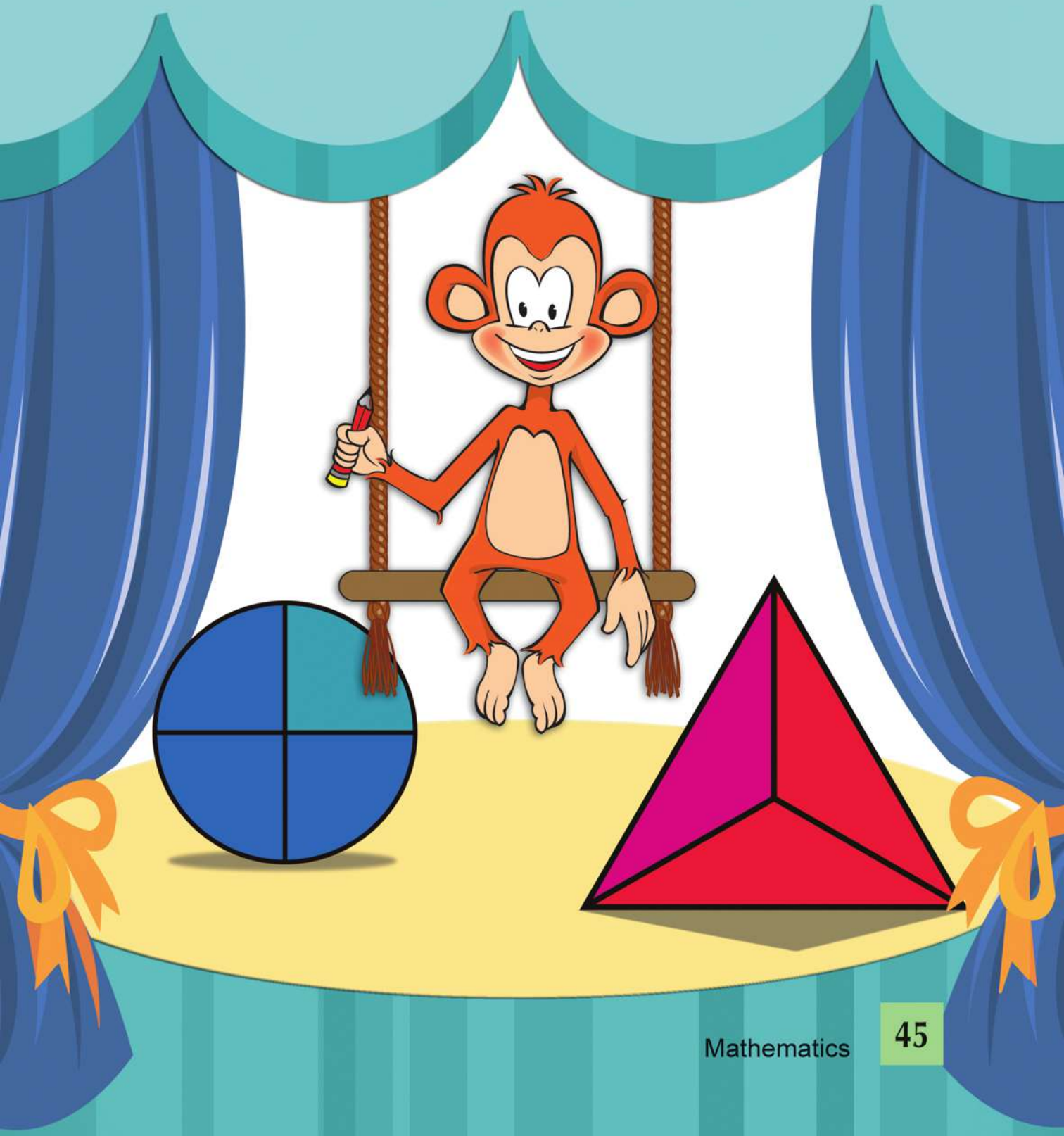
Shape	1	2	3	4	5	6	7	the large square
Its area	1							



- (b) Write the number of:
 Two shapes congruent: and
 Another two shapes that are congruent : and
 Two shapes equal in area but not congruent: and
 Another two shapes equal in area but not congruent: and

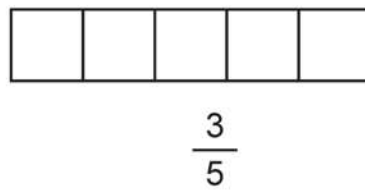
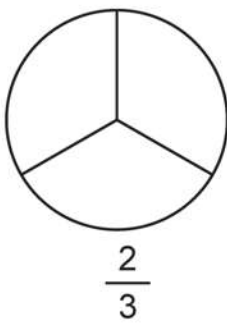
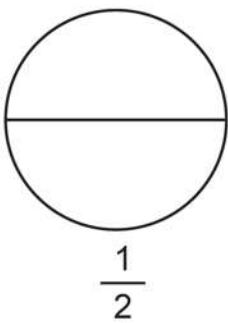
Unit 3

Fractions

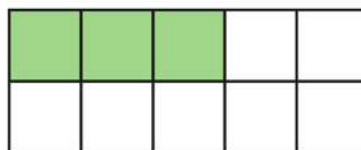
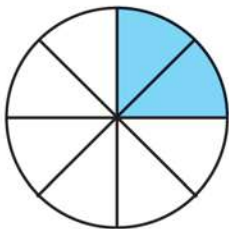
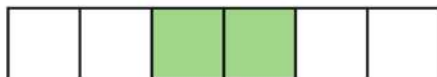
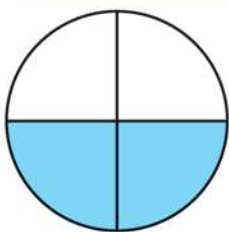


The meaning and Reading of Fractions

(1) Colour according to the fraction:

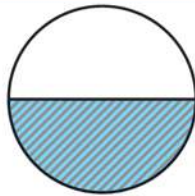


(2) Write the fraction according to the coloured part:



(3) Notice and complete

(a)

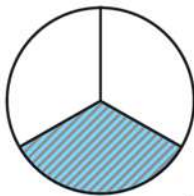


half = $\frac{1}{2}$, **1** = $\frac{2}{2}$ = **two halves**

How many halves in the whole one?

.....

(b)

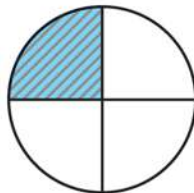


third = $\frac{1}{3}$, **1** = $\frac{3}{3}$

How many thirds in the whole one?

.....

(c)

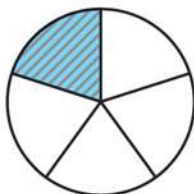


quarter = $\frac{1}{4}$, **1** = $\frac{4}{4}$

How many quarters in the whole one?

.....

(d)



Fifth = $\frac{1}{5}$, **1** = $\frac{5}{5}$

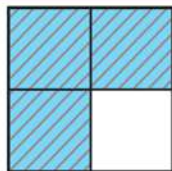
How many fifths in the whole one?

.....

(4) Complete

$$1 = \frac{\dots}{5} = \frac{3}{\dots} = \frac{\dots}{4} = \frac{7}{\dots} = \frac{\dots}{6} = \frac{\dots}{\dots} = \frac{\dots}{\dots}$$

(5) Complete as the example:

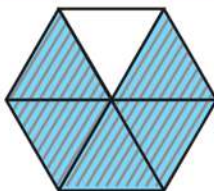


$$\frac{3}{4} = \text{Three quarters}$$



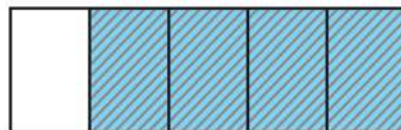
$$\frac{2}{3} = \text{two thirds}$$

(a)



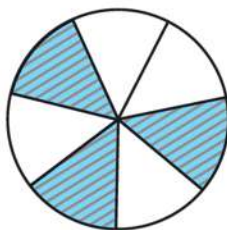
$$\frac{\dots}{\dots} = \dots$$

(b)



$$\frac{\dots}{\dots} = \dots$$

(c)



$$\frac{\dots}{\dots} = \dots$$

(d)



$$\frac{\dots}{\dots} = \dots$$

(6) Write the following fractions:

two thirds = $\frac{\dots}{\dots}$

a half = $\frac{\dots}{\dots}$

four sevenths = $\frac{\dots}{\dots}$

an eighth = $\frac{\dots}{\dots}$

seven eighths = $\frac{\dots}{\dots}$

five sevenths = $\frac{\dots}{\dots}$

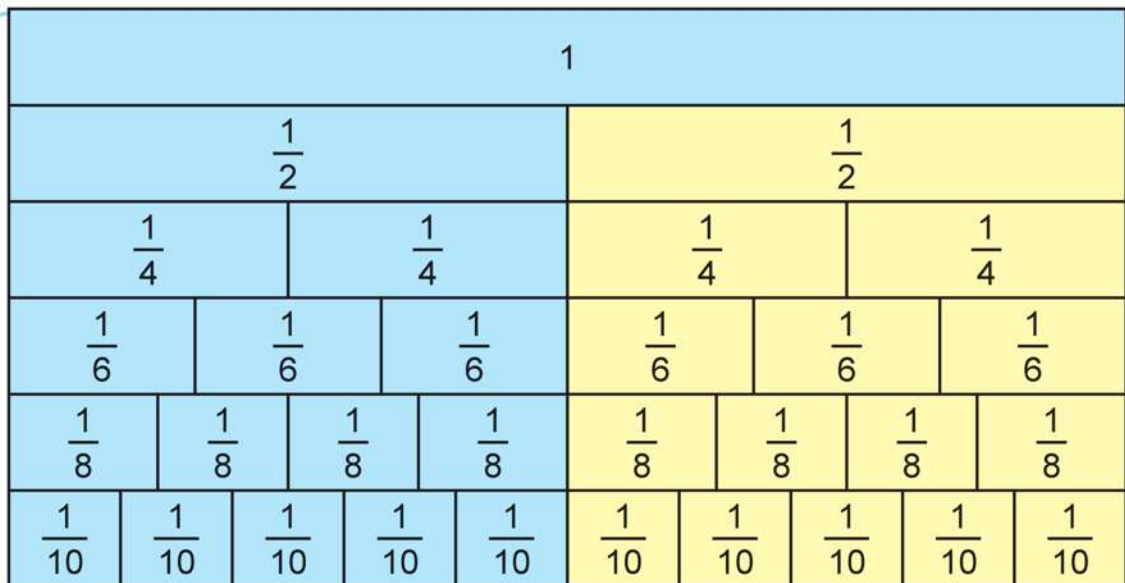
four ninths = $\frac{\dots}{\dots}$

five sixths = $\frac{\dots}{\dots}$

Lesson 2

Equal fractions

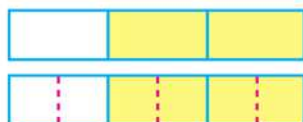
First: Notice the figure:



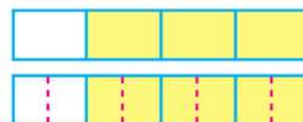
From the figure we notice that:

$$\frac{1}{2} = \frac{2}{4} = \frac{3}{6} = \frac{4}{8} = \frac{5}{10}$$

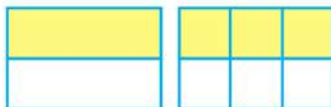
(1) Complete with the help of the figure:



$$\frac{2}{3} = \frac{\dots}{6}$$



$$\frac{3}{4} = \frac{6}{\dots}$$



$$\frac{\dots}{\dots} = \frac{3}{\dots}$$



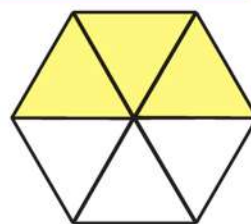
$$\frac{1}{\dots} = \frac{\dots}{\dots}$$

(2) Deduce and complete:

$$\frac{1}{2} = \frac{3}{6} = \frac{5}{10} = \frac{\dots}{12} = \frac{\dots}{14} = \frac{8}{\dots} = \frac{10}{\dots} = \frac{\dots}{\dots}$$

Second: simplifying the fractions

- (1) From the opposite shape we can consider that the coloured part is $\frac{3}{6}$ of the whole figure and we can consider it as $\frac{1}{2}$ of the whole figure as well



So we deduce that: $\frac{3}{6} = \frac{1}{2}$

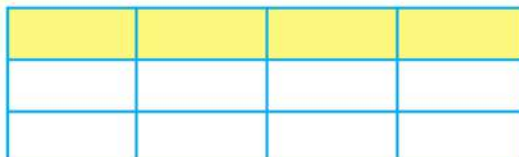
Then complete::

$3 \div \boxed{3} = \dots$, $6 \div \boxed{3} = \dots$

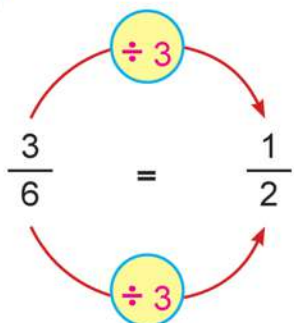
(2) From the opposite shape we can deduce that:

$$\frac{4}{12} = \frac{1}{3}$$

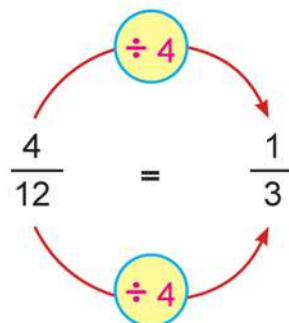
$4 \div \boxed{\dots} = 1$, $12 \div \boxed{\dots} = 3$



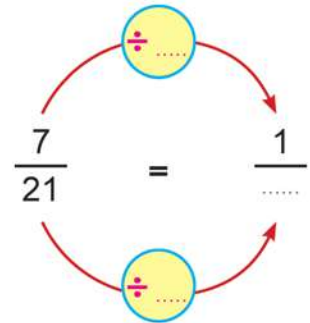
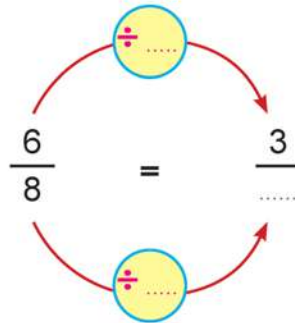
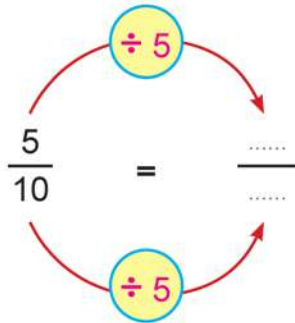
(3)



=



(4) Complete:



(5) Complete:

$$\frac{9}{27} = \frac{1}{\dots}, \quad \frac{15}{25} = \frac{\dots}{5}, \quad \frac{12}{18} = \frac{2}{\dots}, \quad \frac{7}{70} = \frac{1}{\dots}$$

(6) Simplify each of the following fractions:

$$\frac{4}{12} = \frac{\dots}{\dots}, \quad \frac{3}{30} = \frac{\dots}{\dots}, \quad \frac{6}{21} = \frac{\dots}{\dots}, \quad \frac{35}{45} = \frac{\dots}{\dots}$$

(7) Match equal fractions:

$$\frac{3}{4}$$

$$\frac{1}{2}$$

$$\frac{14}{21}$$

$$\frac{4}{14}$$

$$\frac{7}{14}$$

$$\frac{15}{20}$$

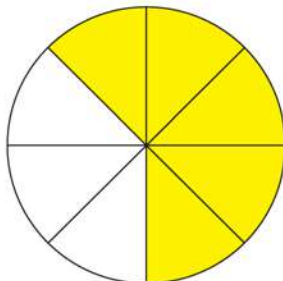
$$\frac{2}{7}$$

$$\frac{2}{3}$$

Comparing and ordering fractions

First: Comparing two fractions:

- (1) Mohamed bought a pie divided into 8 equal pieces. He ate 5 pieces and gave his friend the remaining three pieces.



Complete:

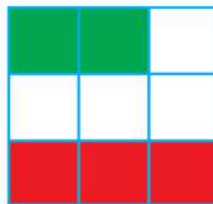
Mohamed ate $\frac{5}{8}$ of the pie and his friend took $\frac{3}{8}$ of the pie.

Which greater: Mohamed's share or his friend's share?

Complete: $\frac{5}{8} > \frac{3}{8}$

- (2) Notice the shape and complete:

$$\frac{4}{9} > \frac{5}{9}$$



- (3) Use the signs (< , = or >) to compare:

$$\frac{3}{4} \square \frac{1}{4}$$

$$\frac{5}{7} \square \frac{6}{7}$$

$$\frac{2}{9} \square \frac{7}{9}$$

$$\frac{4}{5} \square 1$$

$$\frac{3}{3} \square 1$$

$$\frac{2}{5} \square \frac{3}{5}$$

$$\frac{9}{11} \square \frac{7}{11}$$

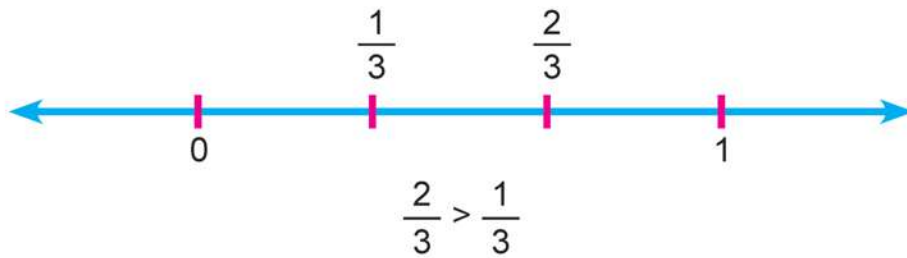
$$\frac{4}{8} \square \frac{7}{8}$$

$$1 \square \frac{6}{6}$$

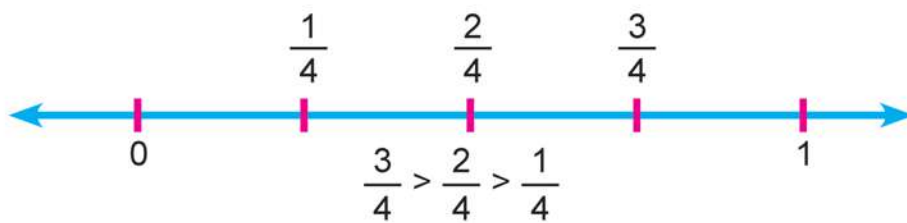
Second: ordering fractions

(1) Notice then complete

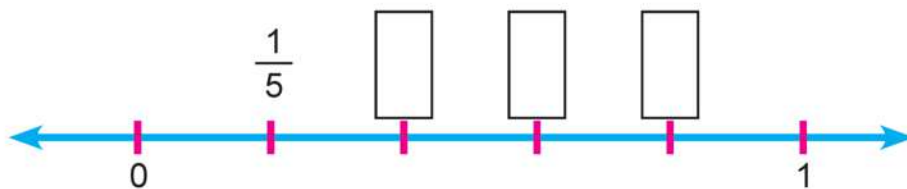
(a)



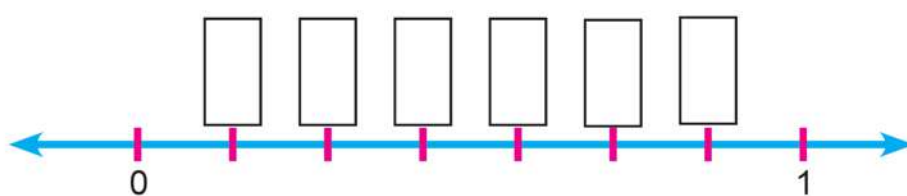
(b)



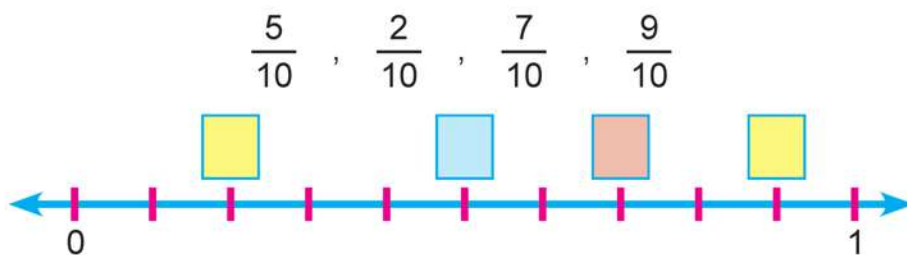
(c)



(d)



(2) Write the following fractions in suitable on the number line:



(3) Arrange the following fractions in ascending order:

(a) $\frac{4}{6}, \frac{1}{6}, \frac{2}{6}$

The order,,

(b) $\frac{5}{8}, \frac{7}{8}, 1, \frac{1}{8}$

The order,,,

(4) Arrange the following fractions in a descending order:

(a) $\frac{1}{5}, \frac{3}{5}, \frac{5}{5}$

The order,,

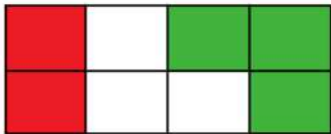
(b) $\frac{2}{9}, \frac{7}{9}, \frac{5}{9}, 1$

The order,,,

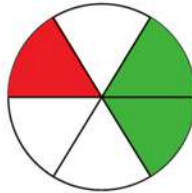
Lesson 4

Adding and subtracting the fractions

(1) Notice each of the following shapes and complete:



$$\frac{2}{8} + \frac{3}{8} = \frac{\dots}{\dots}$$



$$\frac{1}{6} + \frac{2}{6} = \frac{\dots}{\dots}$$



$$\frac{\dots}{\dots} + \frac{\dots}{\dots} = \frac{\dots}{\dots}$$

(2) Add: as in the example:

Example: $\frac{3}{5} + \frac{1}{5} = \frac{4}{5}$



(a) $\frac{2}{7} + \frac{3}{7} = \frac{\dots}{\dots}$

(b) $\frac{1}{9} + \frac{7}{9} = \frac{\dots}{\dots}$

(c) $\frac{1}{4} + \frac{3}{4} = \frac{\dots}{\dots} = \dots$

(3) Complete as in the example:

Example:

$$\frac{2}{9} + \frac{5}{9} = \frac{7}{9} \text{ therefore } \frac{7}{9} - \frac{2}{9} = \frac{5}{9}, \frac{7}{9} - \frac{5}{9} = \frac{2}{9}$$

$$\frac{3}{7} + \frac{1}{7} = \frac{\dots}{\dots}$$

therefore $\frac{\dots}{\dots} - \frac{3}{7} = \frac{1}{7}, \frac{\dots}{\dots} - \frac{\dots}{\dots} = \frac{\dots}{\dots}$

$$\frac{2}{5} + \frac{3}{5} = \frac{\dots}{\dots} = 1$$

therefore $1 - \frac{2}{5} = \frac{\dots}{\dots}, 1 - \frac{\dots}{\dots} = \frac{\dots}{\dots}$

(4) Add

$$\frac{4}{6} + \frac{1}{6} = \frac{\dots}{\dots}$$

;

$$\frac{3}{7} + \frac{1}{7} = \frac{\dots}{\dots}$$

$$\frac{5}{12} + \frac{3}{12} = \frac{\dots}{\dots}$$

;

$$\frac{5}{9} + \frac{4}{9} = \frac{\dots}{\dots} = \dots$$

$$\frac{1}{2} + \frac{1}{2} = \frac{\dots}{\dots} = \dots$$

(5) Subtract:

$$\frac{5}{8} - \frac{3}{8} = \frac{\dots}{\dots}$$

;

$$\frac{4}{9} - \frac{2}{9} = \frac{\dots}{\dots}$$

;

$$\frac{3}{11} - \frac{1}{11} = \frac{\dots}{\dots}$$

$$\frac{5}{6} - \frac{4}{6} = \frac{\dots}{\dots}$$

,

$$\frac{4}{5} - \frac{1}{5} = \frac{\dots}{\dots}$$

,

$$1 - \frac{1}{3} = \frac{\dots}{\dots}$$

(6) Complete:

$$\frac{1}{7} + \frac{4}{7} = \frac{\dots}{\dots}$$

,

$$\frac{5}{12} + \frac{\dots}{\dots} = \frac{9}{12}$$

,

$$\frac{\dots}{\dots} - \frac{3}{10} = 1$$

$$1 - \frac{1}{2} = \frac{\dots}{\dots}$$

,

$$\frac{3}{5} - \frac{\dots}{\dots} = \frac{2}{5}$$

,

$$\frac{4}{7} - \frac{\dots}{\dots} = \frac{1}{7}$$

$$\frac{\dots}{\dots} - \frac{3}{4} = \frac{1}{4}$$

,

$$\frac{\dots}{\dots} + \frac{1}{3} = \frac{2}{3}$$

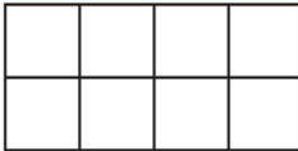
,

$$\frac{\dots}{\dots} - \frac{1}{6} = \frac{1}{6}$$

Exercises

Unit 3

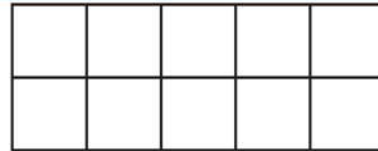
(1) Colour the parts that represent the fraction written under each shape:



$$\frac{5}{8}$$

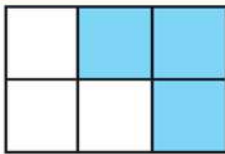


$$\frac{2}{3}$$

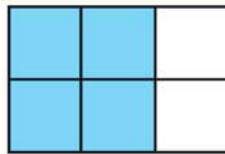


$$\frac{2}{5}$$

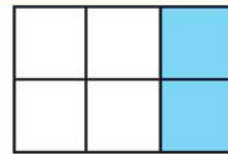
(2) Circle the fraction that represents the coloured part in each of the following shapes:



$$\frac{1}{3}, \frac{1}{6}, \frac{1}{4}, \frac{1}{2}$$



$$\frac{3}{6}, \frac{2}{3}, \frac{1}{4}, \frac{3}{4}$$



$$\frac{1}{3}, \frac{2}{3}, \frac{2}{4}, \frac{3}{4}$$

(3) Complete:

$$\frac{1}{5} + \frac{2}{5} = \frac{\dots}{\dots}, \quad \frac{2}{7} = \frac{6}{\dots}, \quad \frac{8}{9} - \frac{5}{9} = \dots$$

$$\frac{6}{10} = \frac{3}{\dots}, \quad 1 - \frac{5}{8} = \dots, \quad \frac{6}{16} = \frac{\dots}{2}$$

(4) Circle what each of the following fractions equals:

(a) $\frac{3}{5}$ $(\frac{1}{5} + \frac{3}{5}, \frac{6}{20}, 1 - \frac{2}{5})$

(b) $\frac{2}{3}$ $(\frac{6}{9}, \frac{9}{11}, \frac{9}{15})$

(c) $\frac{6}{7}$ $(\frac{3}{7} + \frac{3}{7}, \frac{9}{14}, \frac{12}{15})$

(5) Complete using one of the signs < , = or >:

$$\frac{5}{8} \quad \square \quad \frac{7}{8}$$

$$\frac{2}{3} \quad \square \quad 1$$

$$\frac{11}{13} \quad \square \quad \frac{7}{13}$$

$$1 \quad \square \quad \frac{7}{7}$$

(6) Order the following fractions ascendingly and descendingly:

$$\frac{1}{10}, \quad \frac{3}{10}, \quad \frac{2}{10}, \quad \frac{9}{10}$$

Ascending order:,,,

Descending order:,,,

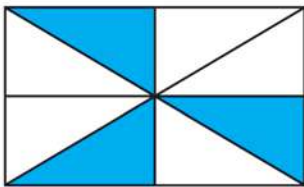
(7) Choose the correct answer

(a) $\frac{15}{25} = \frac{\dots}{5}$ (3, 5, 7)

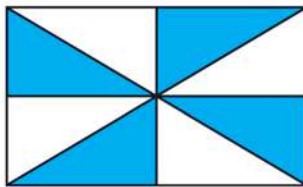
(b) Which the following fraction whose represent the whole one
 ($\frac{4}{4}$, $\frac{4}{3}$, $\frac{1}{4}$)

(c) $\frac{5}{9}$  $\frac{6}{9}$ (> , < , =)

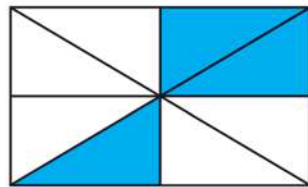
(8) Which of the following shapes represent half?



(a)



(b)



(c)

Activities

Unit 3

(1) Complete the following addition table::

+	$\frac{1}{7}$	$\frac{4}{7}$	$\frac{3}{7}$
$\frac{2}{7}$	$\frac{3}{7}$		
$\frac{1}{7}$			
.....		1	

(2)

- (a) What is the fraction that gives the result $\frac{3}{5}$ if $\frac{2}{5}$ is added to it?
- (b) What is the fraction that gives one whole if $\frac{3}{4}$ is added to it?
- (c) What is the fraction that gives the result $\frac{3}{9}$ if subtracted from $\frac{4}{9}$?
- (d) What is the fraction that gives $\frac{3}{9}$ if $\frac{4}{9}$ is subtracted from it?

(3) Express the following in fractions:

- (a) A box of cheese contains 8 equal pieces - what is each piece in relation to the whole box?
.....
- (b) A pie was divided equally among four friends- what is each one's share?
.....
- (c) A piece of land was divided into 5 equal pieces. One of the pieces was planted cotton. Two pieces were planted wheat and the other two pieces were planted rice. What does each of the following represent in relation to the original piece of land:
Land planted cotton :
Land planted wheat:
Land planted rice:
- (d) A class has 36 pupils. 8 pupils went on a trip what is the fraction that represents the number of pupils that went on the trip in relation to the number of pupils in the class?

$$\frac{3}{8}$$

$$\frac{2}{9}$$

$$\frac{1}{18}$$

$$\frac{1}{4}$$

(4) Complete drawing the arrows so that each arrow goes from the smallest to the greatest:

$$0$$

$$\frac{3}{10}$$

$$\frac{1}{2}$$

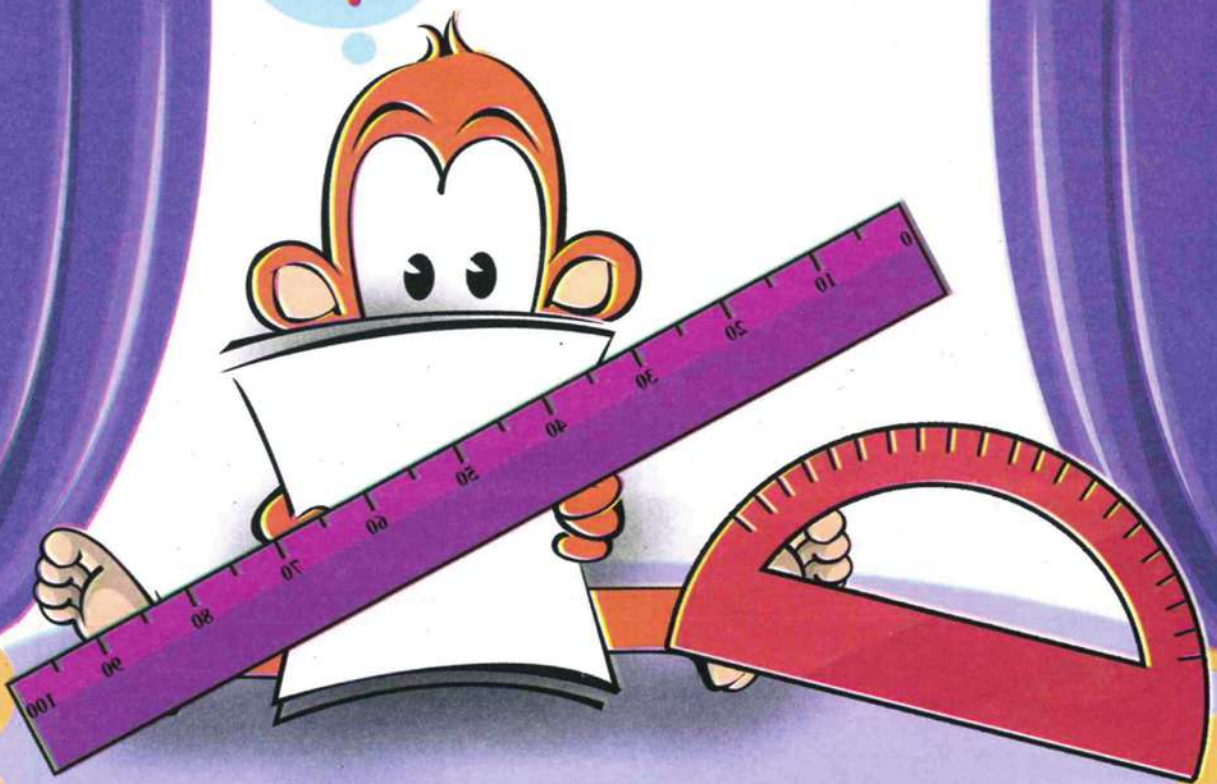
$$\frac{2}{10}$$

$$\frac{9}{10}$$

$$1$$

Unit 4

Measurement



Measuring temperature

(1) Centigrades:

(The thermometer)

How much do you know:

For what do we use the thermometer?

How is the thermometer used?

What is the normal human temperature?

What are the unit's used to measure temperature?

Mention some of the symptoms of human illness.

Complete:

Human temperature is measured by the use of

Temperature is usually estimated by units called



(2) Weather conditions:

Television and other media broadcast the weather report several times every day.

The weather report includes higher and lower temperatures in the capital cities of Egypt's governorates as well as some Arab and world capitals.



Besides temperatures, the weather report also includes a forecast of rains, wind, the ebb and flow (of sea waves), and



other weather conditions. we now move to what

came under "weather conditions" in a cairo

daily newspaper on a day in December:

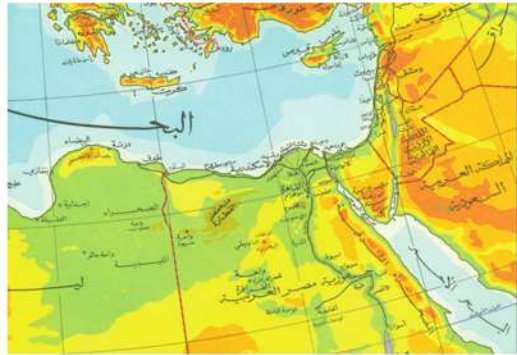


"The weather is slightly warm during the day and cold at

night. Vision decreases in the early morning. Fog in the Delta and Cairo.

Low and medium clouds will hang over the northern and eastern parts of the

country. Frost will accumulate in the inner parts of Sinai. Winds are mostly Northern to North - eastern and mild to moderate. Following are temperatures of some Egyptian cities as well as some Arab and world capitals :



City	Higher Temperatures	Lower Temperatures
Cairo	22°	11°
Alexandria	21°	12°
Aswan	26°	11°
Port Said	21°	16°
Suez	23°	11°
Minya	23°	8°
Ras Sidr	22°	13°
the new Valley	25°	11°
Halayeb & Shalateen	25°	16°
Meccah	30°	16°
Amman	15°	4°
Baghdad	18°	7°
Algeria	20°	9°
New York	11°	5°
Rome	15°	1°
Madrid	10°	1°
London	12°	4°

Unit four

Complete with the help of the previous table :

(a) In Egypt, the highest “higher temperature” was and the lowest was The highest “lower temperature” was and the lowest was

(b) If you wanted to go on a trip in Egypt in December, where would you prefer to go according to the temperatures?

.....

(c) Among the Arab capitals mentioned in the table:

The highest “higher temperature” wasand the lowest was

The highest “lower temperature” wasand the lowest was

(d) Among world capitals :

The highest “higher temperature” wasand the lowest was

The highest “lower temperature” wasand the lowest was

Lesson 2

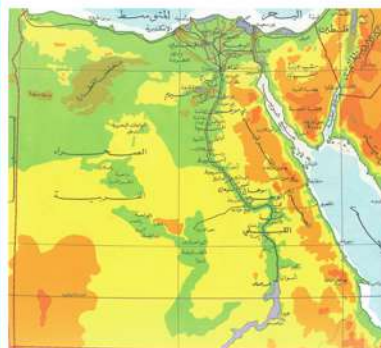
Measuring the Length

It is not preferred to use the units of length we studied before (the metre and the centimeter) in some cases such as calculating distances between cities. This is because these units are too small. Therefore another larger unit has been agreed upon

1kilometre = 1000 metres and is abbreviated as 1 km = 1000m

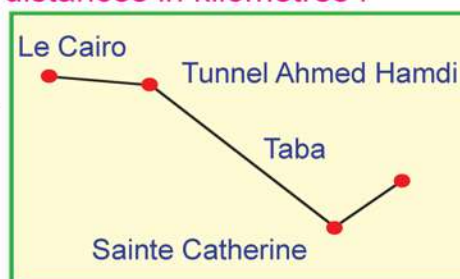
(1) The following table shows the distance in kilometres between a number of cities and locations in the Arab Republic of Egypt.

Between Cities / locations	Distance in kilometres
Cairo - Taba (St. catherine Road)	623
Cairo - Ahmed Hamdy Tunnel	131
Ahmed Hamdy tunnel - St. Catherine	319



With the help of this table, find the following distances in kilometres :

- (a)** The distance between the Ahmed Hamdy tunnel and Taba is :
- (b)** The distance between Cairo and St Catherine to:
- (c)** The distance between St. Catherine and Taba is.:
- (d)** Put the distances you obtained in descending order :



(2) Answer the following questions :

- (a)** In which part of Egypt do St. Catherine and Taba lie?
- (b)** (For oral discussion) Mention what you know about the history of Sinai and Taba in particular.

Unit four

(3) Change from kilometers to metres:

(a) 5 kilometers 5 km = m

(b) 15 kilometers 15 km = m

(c) 12 kilometers 12 km = m

(4) Change from metres to centimetres :

(a) 25 metres

25 m = cm

(b) 43 metres

43 m = cm

(c) 570 metres

570 m = cm

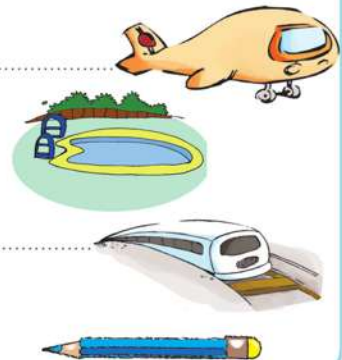
(5) Choose a suitable unit of length (centimeter - metre - kilo metre) for each of the following measurements :

(a) The distance between Cairo and Amman

(b) The length of the swimming pool

(c) The distance between Cairo and Alexandria

(d) The length of a pencil



(6) What is your estimation for the following lengths? Underline the answer you find suitable :

The perimeter of the playground : (100 km , 100 m , 1 km)

The distance between two Egyptian cities : (10000 km , 3000 km , 200 km)

The length of a school notebook : (25 cm , 1m , 1 km)

The height of a child : (140 cm , 2 m , 1 km)

Lesson 3

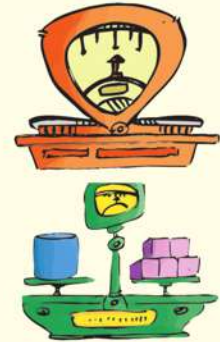
Measuring weight

At the greengrocer's, the grocer's the butcher's, the bakery, and others, the unit of weight used is usually greater than the gram. It is the kilogram.

1 kilogram = 1000 grams

The scales shown in the picture is divided into grams. The maximum weight it can bear (without adding any other weights on it) is kilogram .

Can you use these scales to weigh something greater than a kilogram (3 kilograms) for example? How?



(1) Sawsan went to a shop that sells vegetables and fruits, She bought the following :

1 kgm of tomatoes.

1 kgm of potatoes.

1 kgm of cucumbers.

2 kgm of oranges.

1 kgm of peas.

What is the total sum she paid?

.....

.....

.....

Price list	
items	price /kg m
tomatoes	L.E. 3
green peppers	L.E. 2
peas	L.E. 4
marrow	L.E. 3
cucumber	L.E. 2
onions	L.E. 3
poatatoes	L.E. 3
bananas	L.E. 4
oranges	L.E. 2
apples	L.E. 9
guavas	L.E. 3

(2) Change the following into grams :

(a) 5 k gm

.....

(b) 23 k gm

.....

(c) 18 k gm

.....

Unit four

(3) Choose a suitable unit of weight (kilogram or gram) for each of the following measurements :



pastry



a bottle of medicine



Jewellery



meat

(4) What is your estimation of the following weights? Underline the answers you find suitable :

The weight of a loaf of bread (150 gm , $\frac{1}{2}$ kgm)

The weight of an elephant (10000 gm , 300 kgm)

The weight of jewellery (20 gm , 10 kgm)

The weight of meat (20 gm , 2 kgm)

The weight 20 olives (400 gm , 7 kgm)

Discuss your suggestion for rationing

(5) Complete :

3 kgm = gm , 4500 gm =kgm andgm

4000 gm =kgm , gm = 3 kgm and 30 gm

(6) If the price of a gram of silver is 9 pounds, how much is a silver pot that weight 2 kgm?

Lesson 4

Measuring Time

First: Telling the time :

You have studied some of telling time in the last your :



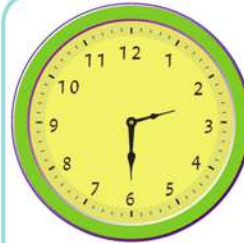
it's quarter past
ten



it's quarter past
Eight



it's half past
eleven



it's half past
two

Notice the position of the minutes's hand and how to read it in each time



past 5
minutes



past 10
minutes



past 25
minutes



25 minutes to



past third



to quarter



to 10
minutes



to 5 minutes



to third

Unit four

Notice the position of both hand in each time :



It's 5 past
three



Its 10 past
seven



It's 5 to nine



It's 10 to twelve

Write the time (Notice the hands of the watch):



It's



It's



It's



It's

The time can be written using the digital watch as in the example:



It's 20 minutes
to ten

9:40



It's 5 minutes
past 5

5:05



It's 10 minutes
to 8

7:50



It's 25
minutes past 3

3:25

(1) Complete:



it's 10 minutes to two

.....



.....

.....



.....

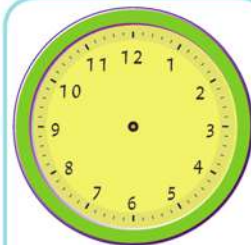
.....



.....

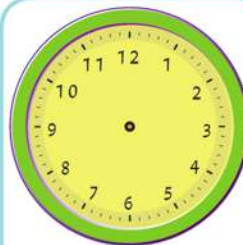
.....

Draw the hands, then complete the missing:



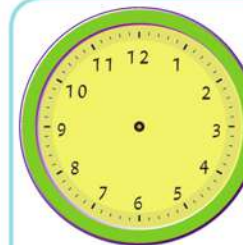
.....

6:50



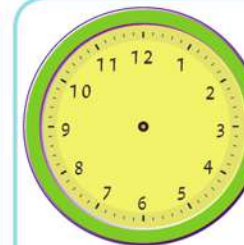
It's 10 minutes past 7

.....



.....

8:55



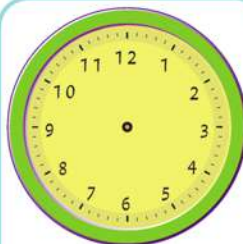
It's third past 4

.....

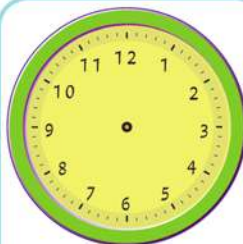
(2) Draw the hands of watch:



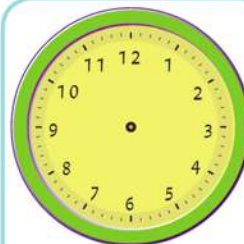
It's 7 o'clock



It's 5 minutes to 9



It's third to 4



It's 25 minutes to 4

Unit four

Second: day, hour and minutes

$$1 \text{ day} = 24 \text{ hours}$$

$$1 \text{ hours} = 60 \text{ minutes}$$

- (1) How many hours are there in two days?
- (2) How many minutes are there in 10 hours?
- (3) How many minutes are there in half an hour?
- (4) How many hours are there in ten days
- (5) How many minutes are there in 4 hours?
- (6) How many minutes are there in quarter of an hour?

(7) Complete:

- (a) 3 hours = minutes (b) a day + 5 hours = hours
(c) 3 days = hour (d) $\frac{1}{3}$ hour = minutes

(8) Arrange the following periods of time ascendingly

- (a) Three hours, 100 minutes, an hour and half,,
(b) Two days, 30 hours, a day and two hours,,
(c) A day, 25 hours, 48 hours,,
(d) 75 minutes, 1 hours, 55 minutes,,

Thirdly: Complete with the help of the calendar in the following figures:

A.D calendar

	January	February	March	April	May	June
Saturday	6 13 20 27	3 10 17 24	31 3 10 17 24	7 14 21 28	5 12 19 26	30 2 9 16 23
Sunday	7 14 21 28	4 11 18 25	4 11 18 25	1 8 15 22 29	6 13 20 27	3 10 17 24
Monday	1 8 15 22 29	5 12 19 26	5 12 19 26	2 9 16 23 30	7 14 21 28	4 11 18 25
Tuesday	2 9 16 23 30	6 13 20 27	6 13 20 27	3 10 17 24	1 8 15 22 29	5 12 19 26
Wednesday	3 10 17 24 31	7 14 21 28	7 14 21 28	4 11 18 25	2 9 16 23 30	6 13 20 27
Thursday	4 11 18 25	1 8 15 22	1 8 15 22 29	5 12 19 26	3 10 17 24 31	7 14 21 28
Friday	5 12 19 26	2 9 16 23	2 9 16 23 30	6 13 20 27	4 11 18 25	1 8 15 22 29
	July	August	September	October	November	December
Saturday	7 14 21 28	4 11 18 25	1 8 15 22 29	6 13 20 27	3 10 17 24	1 8 15 22 29
Sunday	1 8 15 22 29	5 12 19 26	2 9 16 23 30	7 14 21 28	4 11 18 25	2 9 16 23 30
Monday	2 9 16 23 30	6 13 20 27	3 10 17 24	1 8 15 22 29	5 12 19 26	3 10 17 24 31
Tuesday	3 10 17 24 31	7 14 21 28	4 11 18 25	2 9 16 23 30	6 13 20 27	4 11 18 25
Wednesday	4 11 18 25	1 8 15 22 29	5 12 19 26	3 10 17 24 31	7 14 21 28	5 12 19 26
Thursday	5 12 19 26	2 9 16 23 30	6 13 20 27	4 11 18 25	1 8 15 22 29	6 13 20 27
Friday	6 13 20 27	3 10 17 24 31	7 14 21 28	5 12 19 26	2 9 16 23 30	7 14 21 28

(a) The months of the year are :

(b) 1 year = months

(c) The days of the week are :

(d) 1 week = days

(e) The ordinal numbers corresponding the months are :

January (1) . February (2)

(f) The months that have 30 days are:.....

(g) The months that have 31 days are:.....

(h) February from this year has days .

(i) The number of days from this year = (30 ×) + (31 ×) +
= + + = days

(j) What is the day of 7/7 from this year?

(k) What is the data of the first Thursday in December from this year?

.....

Exercises

Unit 4

(1) Complete

4 metres = centimeters , 3 weeks = days

3 kilograms = grams , 1 year and two months = months

2 hours and a quarter = minutes

1 hours and 50 minutes = minutes

(2)

(a) What is the unit used to measure temperature?

(b) What is the normal human temperature?

(c) How many minutes are there in half of an hour?

(3) Arrange ascendingly:

(a) 2400 grams, 250 grams , 1 kilograms

.....

(b) 50 days, 200 hours, 10 days

.....

(c) 3 kilometres , 4000 centimeters , 500 metres , 2000 metres

.....

(4) :

(a) What is the suitable unit for measuring the distance between two countries

(b) What is the suitable unit for calculating the time of a school period?.....

(c) What is the suitable unit for measuring the weight of gold work?

(5) Telling the Time in each of the following :



.....



.....



.....



.....

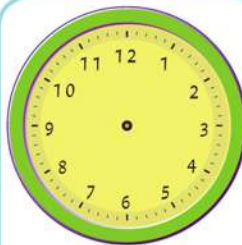
(6) Draw the hands of the watch



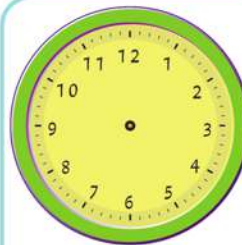
It's 5 minutes
to six



It's 10 minutes
past 4



It's half past 11



It's 25 minutes
to 2

Activities

Unit 4

- (1) (a) What is the agency that studies weather conditions and expectation in Egypt?

..... (Have you visited it on a school trip)

- (b) What kind of clothes would you advise your classmates to wear on the days with the following temperatures..... :

higher	lower	
37 °	30 °	
15 °	10 °	
22 °	17 °	

- (2) (a) What is the birthday of someone who celebrates it only once every 4 years?

- (b) Which is heavier: 10 kilograms of iron or 10 kilograms of cotton?

.....

- (3) How long is the period of time that starts on the beginning of Monday, 15 October, 2009 and ends with the end of Saturday, 27 October 2009?

.....

- (4) A person started a job on the first of March and finished it at the end of August of the same year. How many months did he spend doing this job?

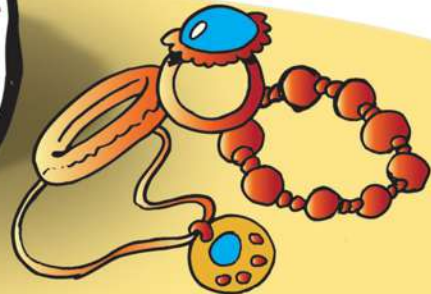
.....

- (5) Medhat walks a distance of 2 kilometres in 20 minutes. How much time does it take him to walk a distance of six kilometres if he walks consistently (with the same speed)? What is the distance he covers in an hour and a half?

.....

Unit 5

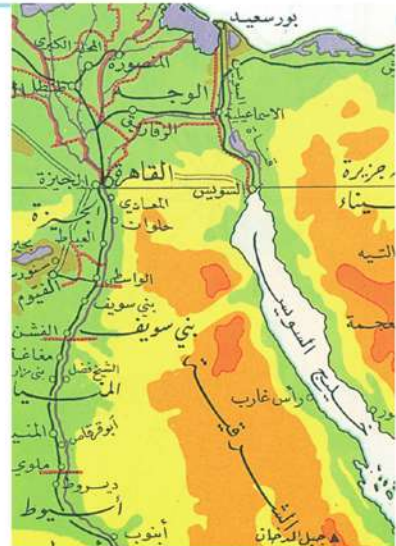
Statistics and probability



Representing Data

- (1) report on the traffic, number of ships and container ships sailing from the Mediterranean sea to the Red Sea found in the records of the Suez Canal Authority was as follows:

Saturday: 30, Sunday: 32, Monday: 45, Tuesday: 55, Wednesday: 37, Thursday: 38, Friday: 52.
Record this data in the following table :

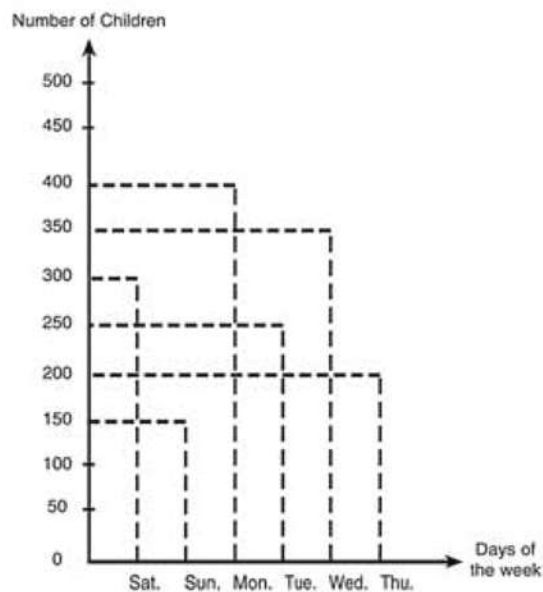


Day	Sat	Sun	Mon	Tue	Wed	Thu	Fri
Number of ships and container ships							

Answer the following questions according to the table:

- (a) Which day had the greatest number of traffic from the Mediterranean Sea to the Red Sea during this week?
What is the number of ships?
- (b) Which day had the smallest number of traffic from the Mediterranean Sea to the Red Sea during this week?
What is the number of ships?
- (c) What is the difference between the greatest and the smallest number of ships and container ships that travelled from the Mediterranean Sea to the Red Sea during that week? - = ships and container ships.
- (d) What are the cities that lie on the Suez Canal?
- (e) (For oral discussion) Tell a historical story about this city.

(2) The following figure shows the number of children vaccinated against polio in one of the week in one of Egypt's governorates, Answer the following questions according to the figure :



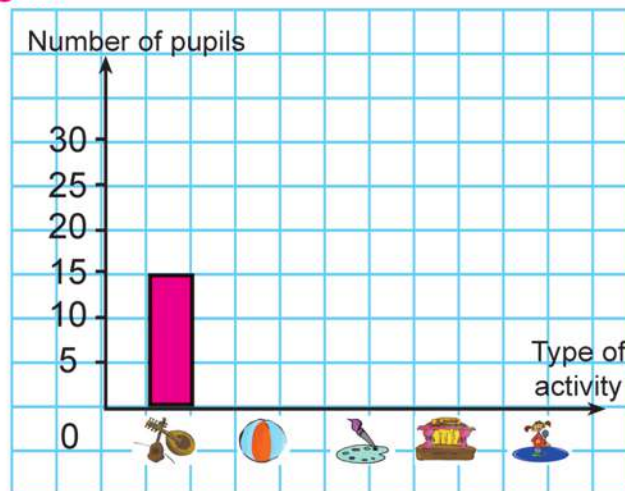
- (a) On what day was the greatest number of children vaccinated?
What is the number of those children?
- (b) On what day was the smallest number of children vaccinated?
What is the number of those children?
- (c) Complete the data in the following table :

Day	Sat.	Sun.	Mon.	Tue.	Wed.	Thu.
Number of children						

(3) The number of the third primary pupils who joined different activity clubs in your school were as follows:

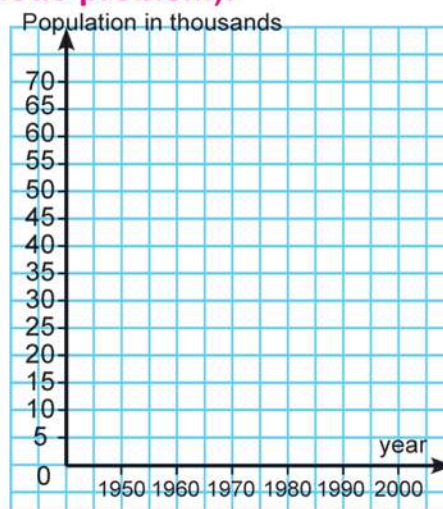
Activity					
Number	15	25	20	15	10

Complete the figure



(4) The following table shows the approximate population of one of the villages in the Arab Republic of Egypt in thousands (approximately) in the given years, complete the graph (with the use of bar-lines as in the previous problem):

Year	Population in thousands
1950	20
1960	25
1970	40
1980	50
1990	55
2000	70



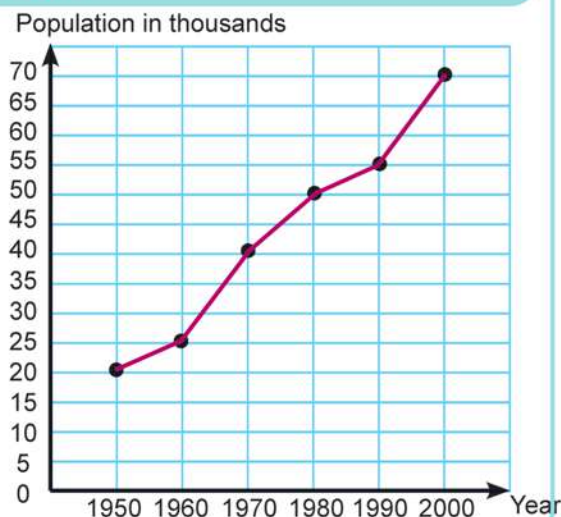
From the previous data answer the following questions:

How much did the population of the village increase from 1950 to 2000?

Representing data with a broken line :

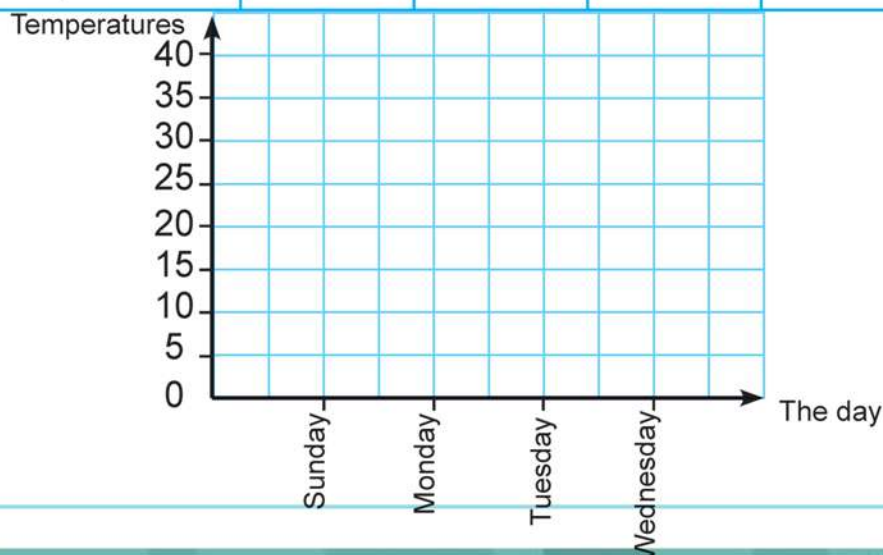
The following figure shows the data in the previous problem represented with a broken line.

Compare the broken line to the columns and deduce how the points on this line were defined.



(5) The following table shows highest temperatures during four days in a city:

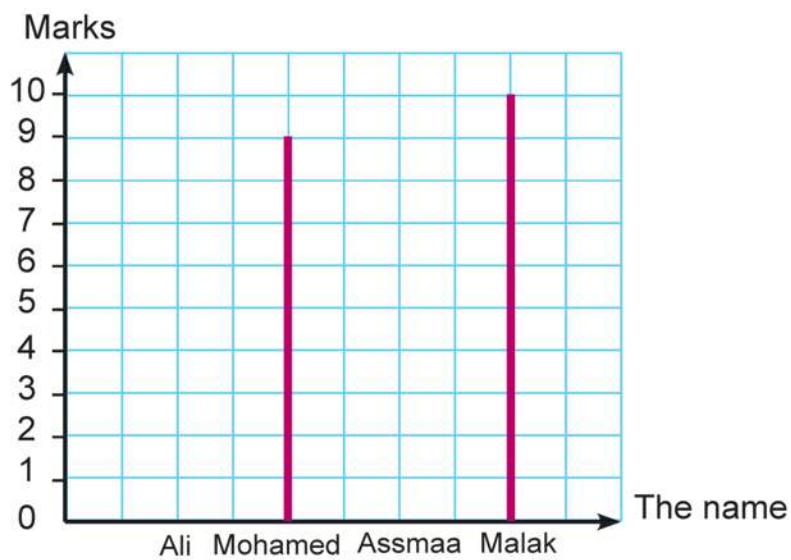
The day	Sunday	Monday	Tuesday	Wednesday
Temperatures	25	30	20	25



(6) The following table shows the marks of 5 pupils in science:

The name	Ali	Mohamed	Assmaa	Malak
Marks	7		8	

Complete the table from graph then complete represent the bar line graph



Lesson 2

Probability

Probability

Certain/ Possible/ Impossible

(1) Complete by writing “certain”, “Possible” or “impossible”:

- (a) It is to rain gold.
- (b) It is that the sun will rise in the morning.
- (c) It is that i will get a high grade in mathematics.
- (d) It is to find a man three metres high.

(2) What do we expect?



Imagine that you closed your eyes and stirred the balls in each container very well and drew one ball out of each.

What do you expect the colour of the drawn ball to be in each case?



(a) The first container:

Complete: ■ It is certain that the ball drawn from the first container will be in colour.

■ It is impossible that it will be in colour



(b) The second container: It is possible that the ball drawn from this container will be in colour

■ It is also possible to be in colour

■ It is impossible to be in colour



(c) The third container:

■ It is certain that the ball drawn from the third container will be in colour

■ It is impossible to be in colour



Guess and expect :

- (1) Assume (as we did previously) that someone closed his eyes and stirred the balls in every container very well and then drew one ball from each container. Can you decide the number of the container from which:



(1)



(2)



(3)

You expect to great extent that the drawn ball will be:

- (a) red (container number)
- (b) yellow (container number)
- (c) Blue (container number)

You expect to a less extent that the drawn ball will be:

- (d) blue (container number)
- (e) red (container number)

- (2) From the previous experiment, decide on the extent of your expectations for each of the following. Write “great”, “moderate”, “weak” or “none”:
- (a) It is possible for the ball drawn from the first container to be blue:
.....
 - (b) It is possible for the ball drawn from the second container to be yellow:
.....
 - (c) It is possible for the ball drawn from the third container to be red:
.....
 - (d) It is possible for the ball drawn from the third container to be blue:
.....

Calculating probability :

If someone closed his eyes and stirred the balls put in the container shown in the opposite figure very well and drew one ball from it, the drawn ball will be either blue or yellow.



- (1) In your opinion is there a greater probability for the drawn ball to be blue or yellow?
- (2) Is there a probability that the drawn ball could be red?

N.B. When an event is impossible we say that the probability of it happening = zero. When it is certain to happen we say that the probability of it happening = 1. If the event, however, is possible the probability of it happening is a fraction (between 0 and 1)

In the previous example :

The probability of the drawn ball being blue is $\frac{1}{5}$

(This is because the number of blue balls = 1 and the whole number of balls = 5).

The probability of the drawn ball being yellow is $= \frac{4}{5}$

(This is because the number of yellow balls = 4 and the whole number of balls = 5)

- (3) Look at the opposite figure, perform the previous experiment and complete



- The probability of the drawn ball being red =
- The probability of the drawn ball being yellow =
- The probability of the drawn ball being blue =

(4) Look at the opposite figure, perform the previous experiment and complete :

- The probability of the drawn ball being blue =
- The probability of the drawn ball being red =



(5) If we flip a coin, we get either heads or tails. Complete

- The probability of getting a head = $\frac{\text{.....}}{\text{.....}}$
- The probability of getting a tail = $\frac{\text{.....}}{\text{.....}}$



(Explain why)?

(6) In a class of 40 pupils, 23 are boys and 17 are girls. One day, one of the pupils was absent

What is the probability of the absent pupil being a boy?

The probability = $\frac{\text{.....}}{\text{.....}}$ (Why?)



Exercises

Unit 5

- (1) The recorded temperatures in one of the weeks from Saturday to the following Friday were as follows: 33° , 31° , 30° , 38° , 34° , 29° and 27° .
record these data in a table and answer the following questions:

- (a) the lowest temperatures was on
- (b) The difference between the highest and lowest temperatures
= - = $^{\circ}$
- (c) The hottest day was
- (d) The temperatures was less than 30° on , ,

- (2) The following table shows the donations of six people to one of the hospitals:

Name of person	Nabila	Aly	Michelle	Sameh	Mervat	Youssef
The amount donated in pounds	50	70	65	55	75	80

Represent these data by bar line graph.

- (3) Choose one of the answers in brackets to express the probability of the occurrence of the following incidents:

- (a) The sun rises in the west (certain- possible - impossible)
- (b) A pupil goes to school (certain- possible - impossible)
- (c) I watch television 4 times a week (certain- possible - impossible)
- (d) I go on a school tripe (certain- possible - impossible)

- (4) What is the probability of seeing an odd number of dots on the face of a piece of dice when you throw it?

Activites

Unit 5

(1) Choose the correct answer from the following answers in brackets about collecting data. Give at least one extra example to support your point of view:

- (a)** Finding the height of a number of plants
(counting and recording - measuring - asking others)
- (b)** The number of cars entering a certain garage
(counting and recording - measuring - asking others)
- (c)** Pupils' preferences in joining different activity clubs
(counting and recording - measuring - asking others)

(2) The following table shows the approximate number of planes that landed or departed from Egypt in five consecutive years (in thousands):

Year	First	Second	Third	Fourth	Fifth
Number of planes in thousands	26	30	35	35	40

- (a)** Represent this data in columns.
- (b)** Mention the names of six airports in Egypt and the names of the governorates in which they are found. (Discuss with your teacher)
- (c)** Mention some of the places you advise Egyptians and tourists to visit in each of the these governorates. (Discuss with your teacher)

(3) Mohamed wrote the names of 7 of his friends on cards as follows :

Said	Ahmed	Khaled	Mina	Salem	Ayman	Amgad
------	-------	--------	------	-------	-------	-------

If we turn these cards, change their position and draw one (without seeing it) the probability that the name written on the drawn card will :

- (a)** begin with «s» is
- (b)** begin with «A» is
- (c)** begin with «W» is
- (d)** have 4 letters is
- (e)** have 5 letters is

Complete:

- (a)** It is certain that the number of letters of the name written on the drawn card is
- (b)** It is impossible that the name written on the drawn card will begin with the letter “ ”.

General Exercise

Unit 1

First questions: Write the following numbers in digits

- 1) Forty five tens =
- 2) ten =
- 3) Seventy =
- 4) Five hundreds and sixty tens =
- 5) Sixty four tens =
- 6) Fifteen tens =
- 7) Ninety eight tens =
- 8) Nine tens =
- 9) Thirty one tens =
- 10) Two tens =
- 11) Three hundreds tens =
- 12) Twenty three hundreds =
- 13) Two hundreds =
- 14) Nine hundreds =

The Second questions :

- 1) If the price of one kg. of orange is 3 pounds. How much is the price of ten kgm

Solution :

The price of the orange = $\times$ = pounds.

- 2) Karol takes 5 pounds from her father every day. Calculate what karol takes in a week?

Solution :

Karol takes in a week = $\times$ = pounds.

- 3) There are two fish`s aquarium, the price of each is 250 pounds. What is the price of two aquariums?

Solution :

the price of two aquariums = $\times$ = pounds.

- 4) Osama bought three boxes of colour, the price of each is ten pounds. Calculate what Osama paid.

Solution :

Osama paid = $\times$ = pounds.

The third questions :

In a shop for saling electric sets, The price of electric sets was shown on it Complete the following :

The type	The number	The price of the unit	Total price
Fridge	20	2000	
Fan	25	200	
hot air set	30	300	
heater	15	200	
blender	10	100	
The sun			

The fourth question: Choose the correct answer from those between brackets:

- | | | | | |
|----|-------------------------|----------------------|------------------------|---------------|
| 1) | 400×2 | <input type="text"/> | 900×10 | (> , < , =) |
| 2) | $3 \times 4 \times 10$ | <input type="text"/> | 15×10 | (> , < , =) |
| 3) | $30 \times 5 \times 2$ | <input type="text"/> | 300 | (> , < , =) |
| 4) | $25 \times 4 \times 20$ | <input type="text"/> | 6000 | (> , < , =) |
| 5) | 700 | <input type="text"/> | $2 \times 35 \times 5$ | (> , < , =) |
| 6) | 3900 | <input type="text"/> | thirty nine tens | (> , < , =) |
| 7) | Two tens + 5 tens | <input type="text"/> | 80 | (> , < , =) |

General Exercises

- 8) $7 \times 20 \times 50$ 6000 ($>$, $<$, $=$)
- 9) eight thousands $7500 + 500$ ($>$, $<$, $=$)
- 10) $2000 + 200$ 4000 ($>$, $<$, $=$)
- 11) 4500 Forty five tens ($>$, $<$, $=$)
- 12) $800 + 200$ Two thousands ($>$, $<$, $=$)

The fifth question:

- 1) $4 \times \dots \times 100 = 12 \times \dots = \dots$
- 2) $6 \times 3 \times 100 = \dots$
- 3) 5 tens + one ten = $\dots$
- 4) $\dots \times \dots \times 1000 = 3000$
- 5) Seven tens = $\dots$
- 6) $2 \times 5 \times 28 = 28 \times \dots = \dots$
- 7) $30 \times 300 = \dots$
- 8) $66 \times 1000 = \dots$ thousands
- 9) $10 \times 1000 = \dots$ and it is read $\dots$
- 10) $4 \times 250 = \dots \times 500$
- 11) 5350 metres = $\dots$ kilometres + 350 metres
- 12) $4 \times 6 \times 10 = \dots \times 10$
- 13) 4 metres = $\dots$ cm

- 14)** $200 + 300 + 500 = \dots\dots\dots$
- 15)** $9 \times \dots\dots\dots = 900$
- 16)** $8000 = 2000 \times \dots\dots\dots$
- 17)** $3 \times 5 = \dots\dots\dots$
- 18)** $2000 = \dots\dots\dots \times 2000$
- 19)** Thirty tens = $\dots\dots\dots$
- 20)** $650 + \dots\dots\dots = 750$
- 21)** $7 \times 200 = 7 \times \dots\dots\dots \times \dots\dots\dots$
- 22)** $3000 = 500 + \dots\dots\dots$
- 23)** Twenty four hundreds = $\dots\dots\dots$

Sixth question: using the numbers

5775 , 4884 , 123 , 5770 , 1221

- a -** The odd numbers are $\dots\dots\dots$
- b -** The even number are $\dots\dots\dots$

Seventh question: Choose the correct answer from the brackets :

- 1)** $804 \div 4 = \dots\dots\dots$ (21 , 201 , 4)
- 2)** $36 \div 6$ $36 \div 4$ ($>$, $<$, $=$)
- 3)** 3003 1001×3 ($>$, $<$, $=$)
- 4)** $25 \div 5$ $25 \div 25$ ($>$, $<$, $=$)

General Exercises

- 5) $8109 \div 9$ 91 ($>$, $<$, $=$)
- 6) $108 \div 2$ 7 ($>$, $<$, $=$)
- 7) $2061 \div 9$ 2061×9 ($>$, $<$, $=$)
- 8) $8080 \div 8 =$ (1010, 110, 101)
- 9) $460 \times 10 =$ (460, 4600, 406)
- 10) $9300 \div 3 =$ (100, 3100, 310)
- 11) $777 \div 7 =$ (11, 111, 101)

Eight question

- a) Bakar bought 164 notebooks, he distributed them equally among his 4 brothers. How many notebooks did each son take?

The share of each brother = = note books

- b) In your school there are 150 pupils in the third primary, they were distributed equally among 3 classes. How many pupils are there in each class?

Number of pupils = $\div$ = pupils

General Exercises

Unit 2

Answer the following questions :

- 1) Find the perimeter of the square whose side length is 3 cm

The perimeter of the square = $\times$ = cm

- 2) Find the perimeter of the triangle whose side lengths are 5 cm , 7cm , and 10 cm

The perimeter of the triangle = + + =
..... cm

- 3) The side lengths of a triangle are equal, each of them equals 7cm
Find the perimeter of the triangle

The perimeter of the triangle = + + = cm

- 4) A rectangle, its length = 5 cm. and its width = 3 cm. Calculate its perimeter.

The perimeter = = cm

- 5) ABC is a triangle where AB = 3 cm. AC = 5 cm. and BC = 4 cm.
Calculate: The perimeter of $\triangle ABC$

The perimeter of $\triangle ABC$ =

- 6) The perimeter of any polygon =

The perimeter of the square = the side length $\times$

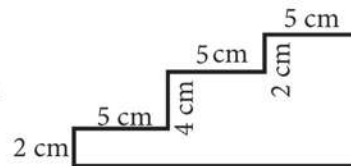
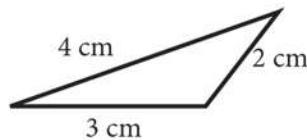
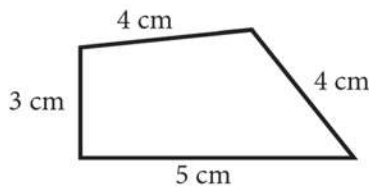
The perimeter of the rectangle = (length +) $\times$

- 7) A triangle piece of land, its perimeter = 100m . If the sum of the length of two sides of it = 70m. Find the length of the third side.

The length of the third side = = m

General Exercises

- 8)** Calculate the perimeter of each of the following shapes using the shown on each shape.:



The perimeter = cm. The perimeter = cm The perimeter = cm

- 9)** A triangular piece of land, the sum of two of its sides equals 90 m, its perimeter is 120 m, Find the length of its third side = =

- (10)** In the opposite figure: Find :

1 - The area of the figure = square units.

2 - The perimeter of the figure = length units

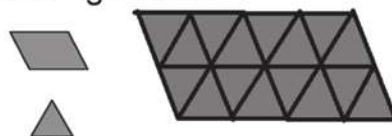


- (11)** In the opposite figure: Find

1 - The perimeter of the figure = length units

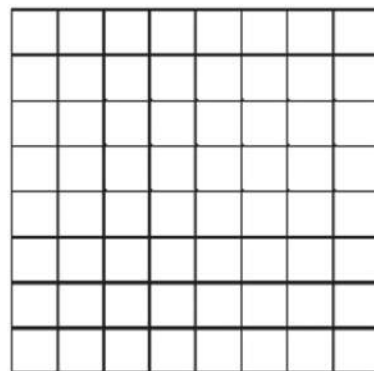
2 - The area of the figure =

3 - The area of the figure =



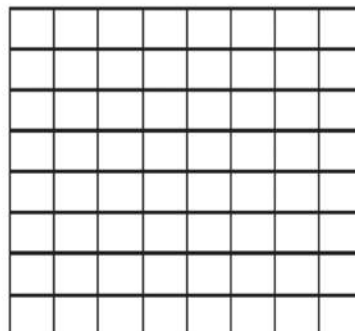
- (12)** On the square lattice

Draw a figure of area 10 square unit .



(13) On the square lattice

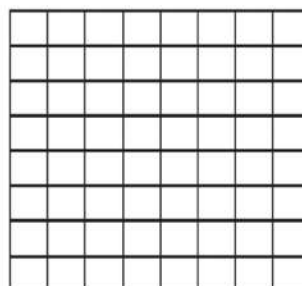
Draw a figure of perimeter 12 length unit.



(14) On the square lattice draw :

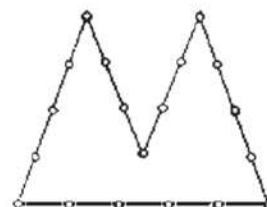
(a) A figure of perimeter 8 length unit.

(b) Another figure of area 9 square units



(15) In the opposite figure: If the distance between any two consecutive points is one centimeter.

Find the perimeter of the figure.



The perimeter of the figure = =

(16) Choose the correct answer from those between brackets :

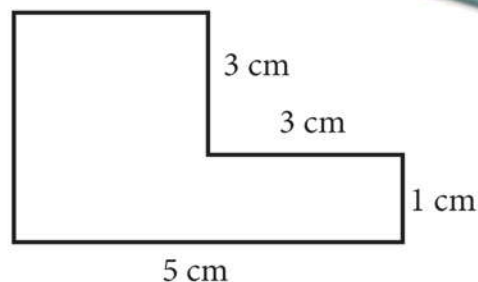
(a) The perimeter of the square whose side length 1cm. is cm
(1, 4, $\frac{1}{4}$)

(b) The perimeter of the rectangle whose length is 4cm. and its width = 2 cm. Its perimeter = cm (12 , 14, 16)

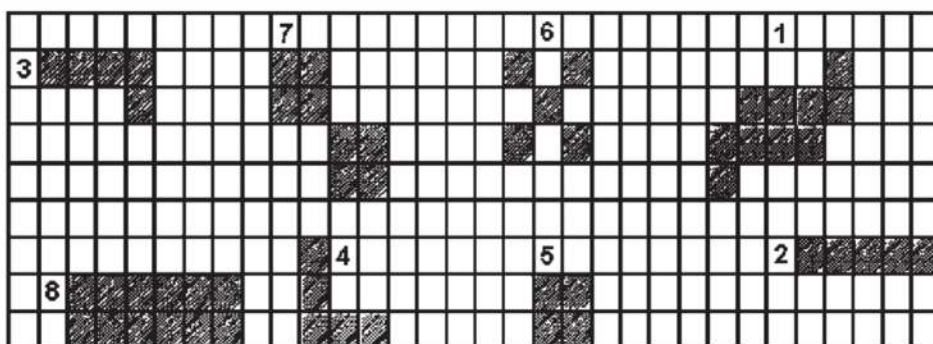
(c) The perimeter of triangle whose sides lengths are 3cm, 4cm and 6cm equals cm. (13 , 14, 15)

General Exercises

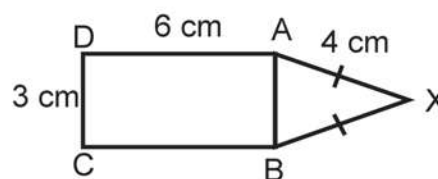
- (17) Calculate the perimeter of the opposite figure.



- (18) In the opposite lattice : Calculate the area of each of the shaded figures then determine the equal areas in square units considering $\square$ as a area unit



- (19) In the opposite figure Find: :



- a) The perimeter of the rectangle ABCD = cm
- b) The perimeter of $\triangle ABX$ = cm
- c) The perimeter of the figure AXBCD = cm

General Exercises

on unit 3

Answer the following questions:

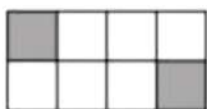
First question: Write the fraction which represents the shaded part in each of the following figures:



The fraction = $\frac{\dots}{\dots}$



The fraction = $\frac{\dots}{\dots}$



The fraction = $\frac{\dots}{\dots}$



The fraction = $\frac{\dots}{\dots}$

Second question: Choose the correct answer from those given between brackets :

- (1) Five sixths = $(\frac{5}{6}, \frac{6}{5}, \frac{2}{6})$
- (2) $\frac{2}{5} + \frac{3}{5} = \dots\dots\dots$ $(\frac{1}{5}, 1, \frac{4}{5})$
- (3) $1 = \frac{6}{\dots\dots\dots}$ $(3, 6, 2)$
- (4) Two seventh + 3 seventh = $(\frac{7}{5}, \frac{5}{7}, \frac{3}{7})$
- (5) $\frac{9}{27} = \frac{1}{\dots\dots\dots}$ $(3, 5, 7)$
- (6) $\frac{5}{7} \square \frac{6}{7}$ $(> , < , =)$
- (7) $\frac{15}{25} = \frac{\dots\dots}{5}$ $(3, 5, 7)$

General Exercises

Third question: Complete the following

(1) Four fifths = $\frac{\dots}{\dots}$

(4) $1 - \frac{3}{4} = \frac{\dots}{\dots}$

(2) $\frac{7}{9} - \frac{5}{9} = \frac{\dots}{9}$

(5) $\frac{5}{7} + \frac{\dots}{\dots} = \frac{6}{7}$

(3) $\frac{3}{4} + \frac{1}{4} = \frac{\dots}{\dots} =$

(6) $\dots - \frac{4}{9} = \frac{3}{9}$

Fourth question : A case of cheese contains 8 equal pieces. Rania ate two pieces. Write fraction which represents what Rana ate relative to the all pieces in the case.

Fifth question: A piece of land divided into 9 equal parts A part of them is planted by cotton, three parts are planted by rice, two parts are planted by wheat.

Write what each of the following represents with respect to the original piece of land .

(1) The part which is planted by cotton = $\dots$

(2) The part which is planted by rice = $\dots$

(3) The part which is planted by wheat = $\dots$

Sixth questions: Use the suitable mark of ($>$, $<$, $=$)

(1) $\frac{3}{5} - \frac{1}{5}$ $\square$ $\frac{3}{5}$

(2) $\frac{7}{9}$ $\square$ $\frac{5}{9} - \frac{2}{9}$

(3) Four Sixths $\square$ $\frac{4}{6}$

(4) $\frac{1}{3}$ $\square$ $1 - \frac{2}{3}$

(5) $\frac{3}{7}$ $\square$ $\frac{2}{7}$

General Exercises

on unit 4

Answer the following questions :-

First questions: Choose the correct answer from those between brackets:

- 1) The distance between Cairo and Alexandria is measured by
(cm , m , km)
- 2) The distance between Cairo and Gidda is measured by
(cm , m , km)
- 3) The suitable unit for measuring the height of a building is... (cm , m , km)
- 4) The length of the swimming pool is measured by (cm , m , km)
- 5) The child's weight of 6 years (age) is
(gm , kg, km)
- 6) The weight of the ring = (14 gm , 4 kg , 14 km)
- 7) The orange's weight = (200 gm , 500 gm , 750 gm)
- 8) The number of the year's days = days (360 , 365 , 375)
- 9) The temperature of the normal person = (35 , 37 , 42)
- 10) The day = hours (7 , 60, 24)
- 11) One day + 5 hours = hours (17 , 24, 29)
- 12) is of length unit (gram, kilogram, metre)
- 13) The unit of measuring weights is (hour , kg , km)

General Exercises

- 14)** The suitable unit for measuring the distance between two countries
(m , kg, km)
- 15)** One year and two months = months (12 , 14, 24)
- 16)** Two hours and a quarter = minutes (115 , 135 , 215)
- 17)** The class period time is measured by
(thermometer , day , minutes)

Second question: Complete

- 1)** 75 metre = $75 \times \dots\dots\dots = \dots\dots\dots$ cm
- 2)** 25 km = $25 \times \dots\dots\dots = \dots\dots\dots$ metre
- 3)** 127 metre = $127 \times \dots\dots\dots = \dots\dots\dots$ cm
- 4)** 17 kilometre = $17 \times \dots\dots\dots = \dots\dots\dots$ m
- 5)** 3 kilograms = $3 \times \dots\dots\dots = \dots\dots\dots$ gm
- 6)** 57 kilograms = $57 \times \dots\dots\dots = \dots\dots\dots$ gm
- 7)** 3450 grams = $\dots\dots\dots$ kg + $\dots\dots\dots$ gm
- 8)** 5 kilograms and 125 grams = $\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$ gm
- 9)** 9 kilograms and quarter of kg = $\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$ gm
- 10)** 6 kilograms and quarter of gram = $\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$ gm
- 11)** 8 kilograms and 375 gm = $\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$ gm
- 12)** One hour and 25 minutes = $\dots\dots\dots + \dots\dots\dots = \dots\dots\dots$ min
- 13)** One hour and minutes = 80 minutes
- 14)** Two hours = minutes

Third question : Put the sign (> , < , =) in the suitable place:

- 1) 250 gm $\frac{1}{4}$ kg
- 2) 4 kg and 50 gm. 450 gm
- 3) One hour and 25 minutes 145 minutes
- 4) 36 hours two days
- 5) 3 days and 5 hours 77 hours.
- 6) One and half a day 37 hours
- 7) One year and 3 months 14 months
- 8) 23 hours One day

Fourth question: Arrange the following measure units, ascendingly and descendingly second time:

- 1) 80 hours , two days , 20 hours

The ascending order is :,,

The descending order is:,,

- 2) Two and half months , 80 days, 48 days

The ascending order is:,,

The descending order is:,,

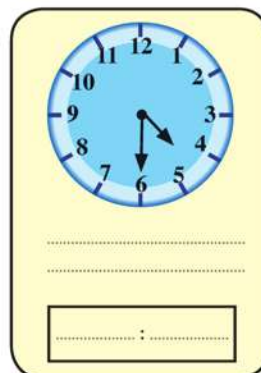
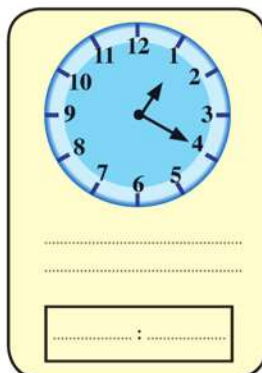
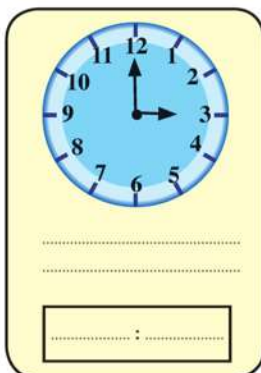
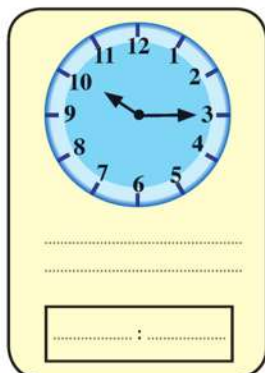
- 3) 3 metres and a quarter of metre , 315 cm. half a metre

The ascending order is:,,

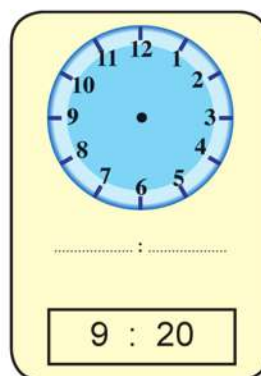
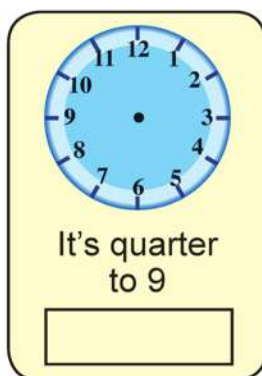
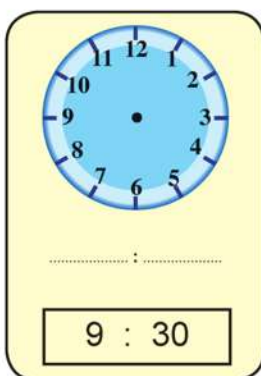
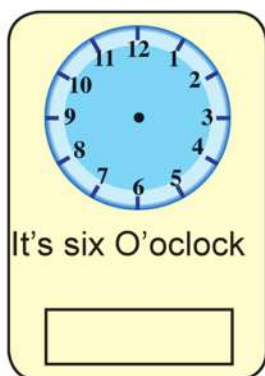
The descending order is:,,

General Exercises

Fifth question: (1) Write the time:



(2) Draw the two hands , then complete:

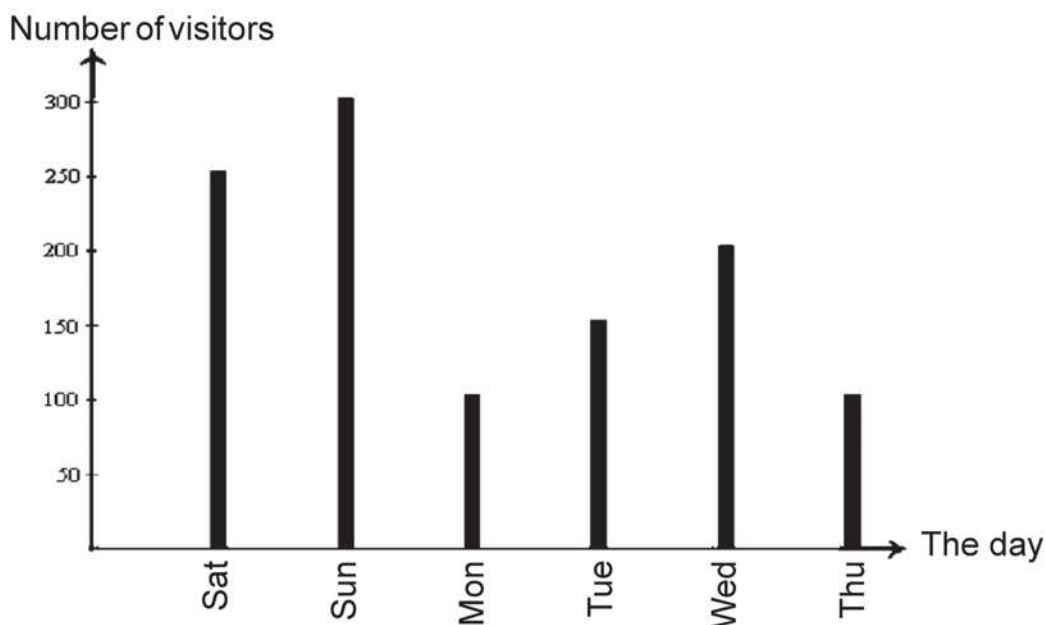


General Exercises

unit 5

(1) First question

The following graph shows the number of visitors to the zoo during 6 days:



From the graph find.

First: complete the following table :

The day	Sat	Sun	Mon	Tue	Wed	Thu
Number of visitors						

Second: Complete:

- 1) The least number of the visitors is on
- 2) The greatest number of the visitors is on
- 3) The difference between the greatest number and the least number of visitors =

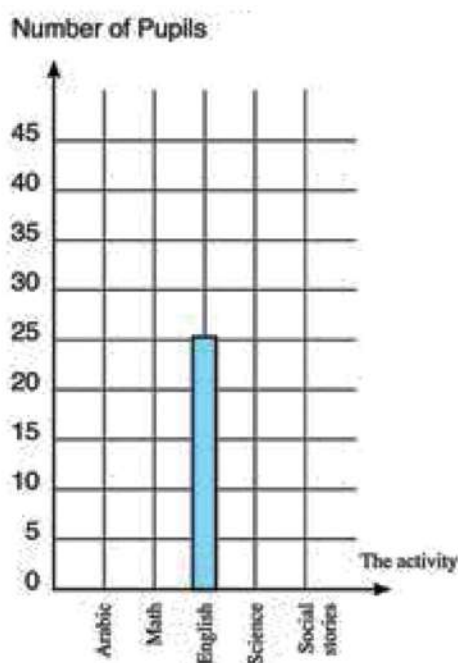
General Exercises

4) The number of visitors who visited the zoo in Monday and Thursday =

The second question :

If the number of the pupils who participated in school activity teams of the different subjects of 5th grade primary in your school due to the due to the general evaluation. Of the fundamental subjects as follows represent the previous data by bar charts :

Activity	Number
Arab.	30
Maths	45
English	
Science	20
Social stories	15



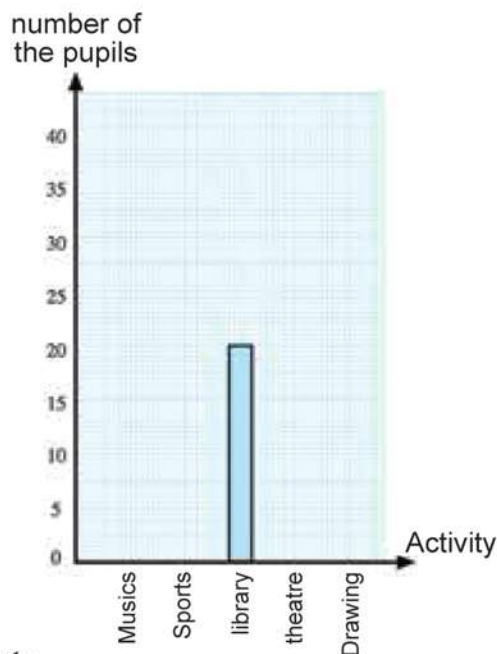
Complete the table and then

Represent these data by bar charts .

The third question :

If the number of the pupils who took part in the school different activities of 3rd grade primary in your school (General evaluation), subjects out the total marks as follows :

The activity	The number
Musics	15
Sports	30
library	
theatre	15
Drawing	10



First: Represent these data by bar charts

Second: Complete the following :

- 1) The least number of the pupils took part in the activity of
- 2) The most of pupils took part in the activity of
- 3) The number of pupils who took part in drawing and library together is
- 4) The number of pupils who took part in musics and sports activities is
- 5) The difference between the pupils who took part in library and theatre activities
- 6) Arrange the activities which the pupils practise them ascending
- 7) Arrange the activities which the pupils practise them descendingly.

General Exercises

The fourth question:

The following table shows the kinds of the electric sets in a shop of electric sets

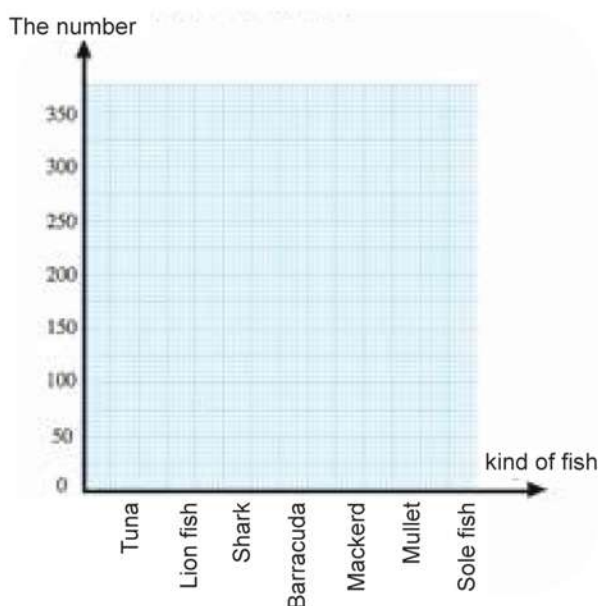
The sets	Fridge	Oven	heater	Fan	Mixture
The number	60	80	70	100	20

Represent these data by bar charts

The fifth question :

The following table shows some of kind of fish which live in red sea

The kind of fish	The number
Tuna	300
Lion fish	100
Shark	20
Barracuda	150
Mackerd	200
Mullet	250
Sole fish	150



First: Represent these data by bar charts

Second complete using the previous table

- 1) The most number of fish is
- 2) The least number of fish is
- 3) There are two kinds of fish having the same number, they are and
- 4) The sum of greatest and least number of fish = +

5) The difference between the greatest and least number of fish -

The sixth question :

The following table shows the production of some crops which some desert governorates are well known by them.

Crops	balm	olives	Almonda	maiz	fruit	vegetables
The number	70	50	60	50	10	15

Represent these data by bar chart

The seventh question :

The following table shows the production of some fundamental crops in the Arab republic of Egypt.

Crops	Sugar cane	Rice	Wheat	Beans	Cotton	Lentils
The number	70	80	65	70	90	40

Represent these data by bar chart

The eighth question:

Complete by writing the word (sure, possible, impossible)

- 1) It is that the sky rains rose
- 2) It is that the sun rises from East
- 3) It is that a man is of length 5 metres.
- 4) It is that the student gets the full mark
- 5) It is that the elephant flies.
- 6) It is that the crocodile lives dry land.
- 7) It is that the hen bear.
- 8) It is that the sky is cloudy.
- 9) It is that the fish live in water

General Exercises

The Ninth question:

When a die is tossed once, and the upper face is observed, Find the following probabilities:

- 1) Appearance of an odd number =
- 2) Appearance of an even number =
- 3) Appearance a number less than 4 =
- 4) Appearance a number greater than 4 =
- 5) Appearance the number 7 =
- 6) Appearance the number 1 , 2, 3, 4, 5 , 6 =

The tenth question:

A box contains 12 balls, 5 balls are white, 4 balls are red, 3 balls are black
Find the probability of each of the following events :

- 1) The drawn ball is red =
- 2) The drawn ball is white =
- 3) The drawn ball is white or black =
- 4) The drawn ball is white or red or black =
- 5) The drawn ball is not red =
- 6) The drawn ball is not white =
- 7) The drawn ball is black =

The eleven question :

As throwing a metallic coin once and observing the upper face. Complete:

- 1) The probability of appearing a head = $\frac{\dots}{\dots}$
- 2) The probability of appearing a tail = $\frac{\dots}{\dots}$
- 3) The probability of appearing a head or a tail =

The twelve question: Choose the correct answer from those between brackets

- 1) As tossing a metallic coin once and observing the upper face then the probability of appearing a head ($\frac{1}{2}$, 1 , zero)
- 2) The sun rises from East is a event (certain, possible, impossible)
- 3) The probability of getting an even number when tossing a die once = ($\frac{1}{4}$, $\frac{1}{2}$, $\frac{1}{6}$)
- 4) The probability of the certain event = ($\frac{1}{2}$, 1 , zero)
- 5) The probability of the impossible event = ($\frac{1}{2}$, 1 , zero)
- 6) The probability of the number 8 when tossing a die once = ($\frac{1}{8}$, 1 , zero)
- 7) The probability of getting a number less than 3 when a die is tossed once = ($\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{6}$)
- 8) The probability of getting a number less than 1 when a die is tossed once = (zero , $\frac{1}{2}$, $\frac{1}{6}$)

Model tests

Model (1)

Answer the following questions :

First : Choose the correct answer:

(1) 5 tens = $5 \times$

a) 10

b) 100

c) 1000

(2) Five sixths $\square \frac{6}{6}$

a) >

b) <

c) =

(3) Which of the following represent an odd number:

a) 5361

b) 5362

c) 5366

(4) Which of the following fractions represent $\frac{1}{4}$:



(5) $\frac{1}{7} + \frac{2}{7} =$

a) 1

b) $\frac{3}{7}$

c) $\frac{1}{7}$

(6) The suitable unit for measuring the length of the pencil is

a) metre

b) centimetre

c) kilometre

(7) The side length of a square its perimeter 20cm = cm

a) 5

b) 80

c) 10

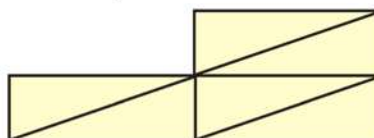
(8) $\frac{1}{2} >$

a) $\frac{3}{6}$

b) $\frac{3}{4}$

c) $\frac{1}{3}$

(9) The area of the figure



=



a) 3

b) 4

c) 6

(10) The number if multiplied by 213 the result will be 21300 is

a) 10

b) 100

c) 1000

(11) A father wants to distribute 183 pieces of chocolate among his 3 his sons, then the share of each son = piece

a) 16

b) 61

c) 26

(12) Ahmed wants to buy 135 notes, if the price of one note is 8 pounds, then the total money of what Ahmed pay requires:

a) adding $135 + 8$

b) multiplying 135×8

c) dividing $135 \div 8$

(13) The fraction if added to $\frac{4}{6}$ the result will be 1 is

a) $\frac{4}{6}$

b) $\frac{2}{6}$

c) $\frac{4}{4}$

(14) Telling the time

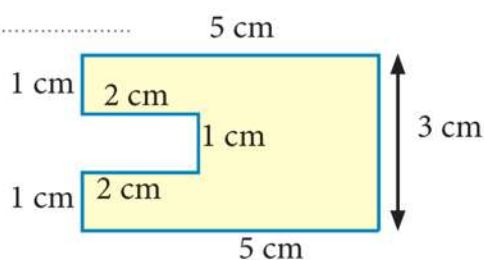


a) 6 o'clock

b) 5 minutes to 6

c) 5 minutes past 6

(15) The perimeter of the opposite figure



a) 10 cm

b) 15cm

c) 20cm

(16) The probability of appearance of an odd number when tossing a die once is :

a) 1

b) half

c) zero

Model test

(17) The probability of the impossible event =

- a) 1 b) half c) zero

(18) Two years and one month = months

- a) 12 b) 24 c) 25

(19) The Probability of getting a head when tossing a coin once =

- a) 0 b) $\frac{1}{2}$ c) 1

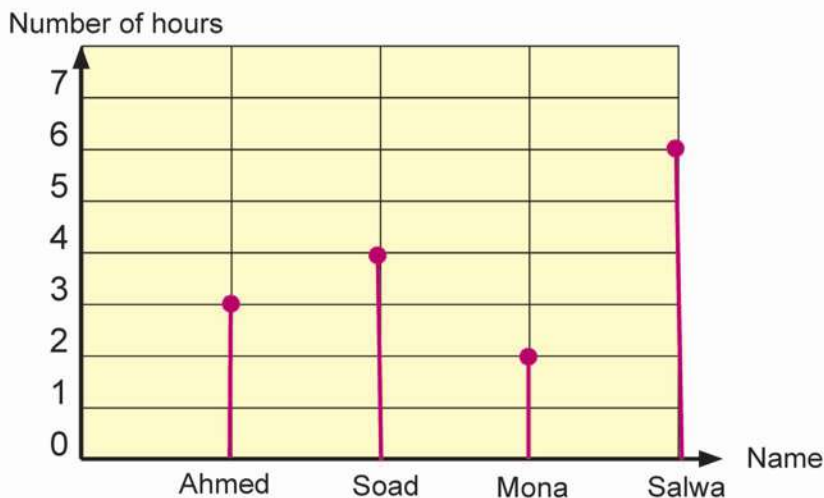
Second: Complete the following :

(1) $\frac{16}{24} = \frac{4}{\dots\dots}$

(2) $624 \div 2 = \dots\dots\dots$

(3) $300 \square 400 - (10 \times 20)$

(4) The opposite figure shows the number of hours of studying for a group of pupils, study the figure, then state the name of the pupil who study the greatest numbers of hours.



(5) $236 \times 4 = \dots\dots\dots$

(6) 4, 40, 400,,,, (the same pattern)

- (7) The Probability of getting an even number when a die is tossed once =

Third:

- (1) Amr bought 4 jackets if the price of each one is

L.E 375 Find what Amr paid

What Amr paid = - L.E

- (2) Calculate the perimeter of the rectangle whose length 8cm, its width 4 cm, the perimeter of the rectangle = = cm

Model tests

Model (2)

Answer the following questions :

First : Choose the correct answer:

(1) $1 - \frac{4}{6} = \frac{1}{6} + \dots\dots\dots$

a) $\frac{1}{6}$

b) $\frac{2}{6}$

c) $\frac{6}{6}$

(2) A box contains 10 symmetric balls 5 white and the rest is red if a ball is drawn randomly, then the probability of the drawn ball is red =

a) $\frac{1}{4}$

b) $\frac{1}{3}$

c) $\frac{1}{2}$

(3) Two days and two hours = hours

a) 40

b) 50

c) 96

(4) Which of the following groups of fractions are arranged in an ascending order:

a) $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{5}$

b) $\frac{1}{2}$, $\frac{1}{3}$, $\frac{1}{5}$, $\frac{1}{4}$

c) $\frac{1}{5}$, $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$

(5) $136 \times 100 = \dots\dots\dots$

a) 360

b) 13600

c) 136000

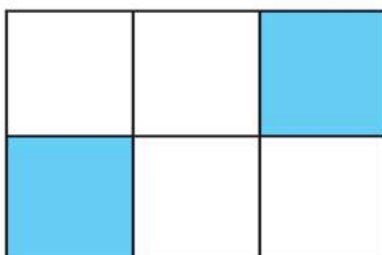
(6) The normal human temperature

a) 73°

b) 37°

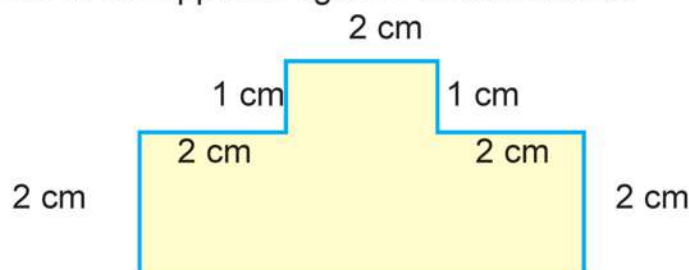
c) 30°

- (7) The fraction which represents the shaded part is



- a) $\frac{2}{8}$ b) $\frac{1}{2}$ c) $\frac{1}{3}$

- (8) The perimeter of the opposite figure = cm



- a) 12 b) 18 c) 20

- (9) A father wants to distribute L.E 206 between his sons mohamed, Ahmed, then the share of each of them = pounds

- a) 102 b) 103 c) 120

- (10) The fraction if added to $\frac{1}{4}$ the result will be $\frac{2}{4}$ is

- a) $\frac{1}{2}$ b) $\frac{1}{3}$ c) $\frac{1}{4}$

Model test

(11) The probability of the certain event is =

a) $\frac{1}{2}$

b) zero

c) 1

(12) $\frac{1}{2} + \frac{1}{2} \square \frac{5}{5}$

a) >

b) =

c) <

(13) Which of the following number is not an even number

a) 268

b) 407

c) 610

(14) 3 tens + = 33

a) 3

b) 9

c) 6

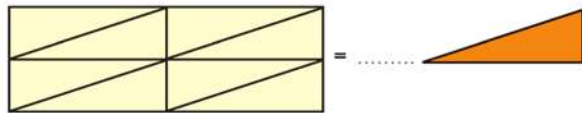
(15) The suitable unit for measuring the length of your class is the

a) metre

b) centimetre

c) kilometre

(16) The area of the opposite figure



a) 4

b) 8

c) 12

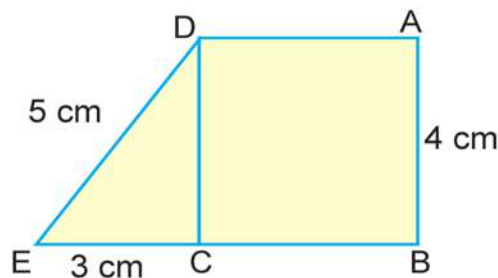
(17) Soha wanted to buy 813 notes for 6 pounds each, then the total price requires operation

a) addition

b) multiplication

c) division

(18) In the opposite figure ABCD is a square, AB = 4cm, DE = 5cm, CE = 3cm then the perimeter of the figure ABED =cm



a) 22

b) 20

c) 24

(19) 6 kg, 250 gm : gm

a) 6250

b) 6500

c) 6750

Second: Complete the following :

- (1) The number that if multiplied by 615 then result will be 615000 is
- (2) The probability of getting atail when tossing a coin once =
- (3) Hossam has 6 banknotes of 100 pounds, and 40 banknotes of 10 pounds, then the total money of what Hossam has = pounds
- (4) 12 , 36, 108 ,, (by the same pattern)
- (5) The following table shows the number of hours that some pupils study, the difference between the greatest, smallest number of hours =

The name	Mona	Ahmed	Salma	Mohamed
Number of hours	6	8	4	5

- (6) The number that if divided by 8 the result will be 16 is
- (7) 3 hundreds 4 hundreds - (10x20)

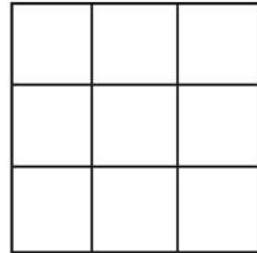
Third: Answer the following:

- (1) Hossam bought 6 shoes if the price of each shoe 95 pounds. How many pounds did he pay?
The price of all shoes = = L.E

Model test

- (2) Calculate the area of the opposite figure

The area =



Model test

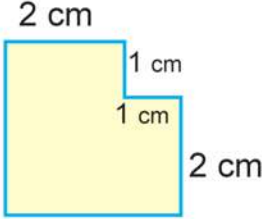
Model

(3)

First: Choose the correct answer:

(1) $\frac{5}{8} = \frac{\dots\dots}{24}$ (13, 14, 15)

(2) Appearing a tail when tossing a coin once is event
(Certain, possible, impossible)

(3) The perimeter of the figure  = cm
(6, 9, 12)

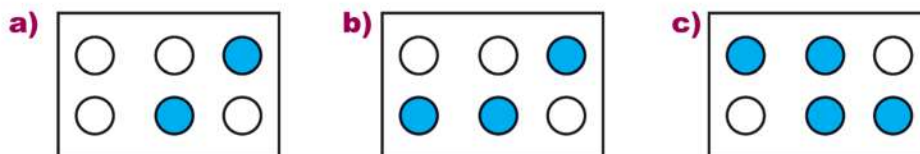
(4) $20 \times \dots\dots\dots = 200$ (1, 10, 100)

(5) Four sixths $\square \frac{1}{6} + \frac{5}{6}$ ($>$ ou $<$ ou $=$)

(6) One year and quarter year = months (12, 14, 15)

(7) The human temperature is measured by the use of
(metre, thermometer, centimetre)

(8) The coloured circled represent half in the figure



(9) $154 \square 100 = 15400$ ($+$, $\times$, $\div$)

(10) The number that multiplied by 5, the result will be 255 is
(5, 51, 15)

Model test

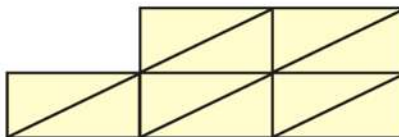
- (11) A bag contains 10 symmetrical balls , 5 of them are red and the rest are white, then the probability of the drawn ball is white is

$(\frac{1}{4}, \frac{1}{3}, \frac{1}{2})$

- (12) One and half hour = minutes

(30 , 90 , 100)

- (13) The area of the figure



= 

(5, 10, 2)

- (14) A teacher bought 402 notes to distribute them among some pupils, if the the price of one note equals 4 pounds, then the total cost requires operation (addition , multiplication , division)

- (15) A man distributed 603 pounds equally among his three sons, then the share of each son = pounds (101, 102 , 201)

- (16) The fraction which added to $\frac{5}{7}$ the result equals a whole unit is

$(\frac{1}{7}, \frac{2}{7}, \frac{7}{7})$

- (17) The telling time of  is

(quarter past seven, thirty five minutes past three, seven o'clock)

- (18) The perimeter of the figure  = cm (6, 9, 10)

- (19) The probability fo the impossible event = $(0, \frac{1}{2}, 1)$

Second: Complete each of the following:

(1) The ascending order of the fractions $\frac{1}{8}, \frac{7}{8}, \frac{5}{8}, \frac{3}{8}$ is,,,

(2) The number that it divided by 6 the result will be 13 is

(3) 6, 12, 24 ,,, (in the same pattern)

(4) The following temperatures recorded in one city during 5 consecutive days as follows:

Day	Saturday	Sunday	Monday	Tuesday	Wednesday	Thursday
Temperatures	30°	29°	32°	39°	36°	31°

The day was the temperature is the highest

(5) From numbers : 6374 , 8651 , 4205 , 1352 complete the odd numbers are,

(6) The opposite figure represents a rectangular piece of land divided into 9 equal 3 parts of them were planted with red flowers, So these



parts represent $\frac{\text{.....}}{\text{.....}}$ with respect to the whole land

(7) $2154 \times 3 = \dots\dots\dots$

Model test

Third: Answer the following questions

- (1) Salwa bought 6 bags, the price of each one is 175 pounds
How much money did she pay?
She paid = = L.E
- (2) A peice of triangular land, the sun of teo sides of it equals 100m,
its perimeter 150m, find the length of its third side
the length of third side =
= m